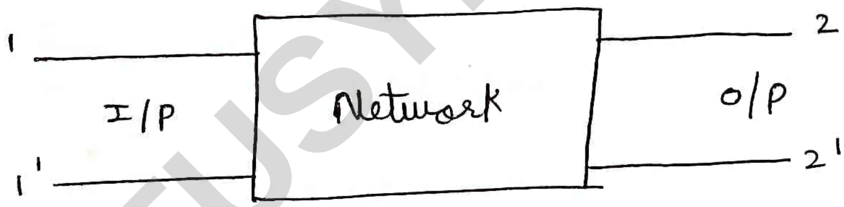
Introduction

A General n/w having 2 pairs of terminals form's an important building block in many system communication, control systems. The pair of terminals at which the electrical energy enters is known as I/P terminal, while a pair of terminal which signal leaves is known as O/P terminal.

A pair of terminals at which A signal may enter or leave a n/w is called port.

A n/w having 2 pairs of terminals is known as 2 port network.



where, 1 & 1' are I/P terminals
2 & 2' are O/P terminals

A two port n/w has 4 variables V_1, V_2, I_1 & I_2 out of these four, two variables can be dependent variable & other two variables are independent.

Depending on which variables are choosen as independent variable 4 sets of parameter are defined for a 2 port network they are.

① open ckt impedence (z- parameters)

Dependent variables $= V_1 V_2$

independent variables $= I_1 I_2$

② short ckt ~~impedence~~ admitance (Y - Parameters)

Dependent variables $= I_1 , I_2$

independent variables $= V_1 , V_2$

③ Hybrid or h - parameter

Dependent variable $= V_1 , I_2$

independent variable $= V_2 , I_1$

④ Transmission / chain / ABCD / General circuit parameters.

Dependent variable: V_1 , I_1

independent variable: V_2 , I_2

-> open ckt impedence or z parameters:

Let V_1 & V_2 be the dependent variables, I_1 & I_2 are independent variables

$$\therefore V_1 = f_1 (I_1 , I_2)$$

$$V_2 = f_2 (I_1 , I_2)$$

The i/p & o/p voltages V_1 & V_2 can be expressed in terms of i/p & o/p currents I_1 & I_2 as

$$V_1 = z_{11} I_1 + z_{12} I_2 \quad \ldots ①$$

$$V_2 = z_{21} I_1 + z_{22} I_2 \quad \ldots ②$$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

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Put $I_2 = 0$ in eq ① & ②

$V_1 = Z_{11} I_1$

$$\boxed{Z_{11} = \frac{V_1}{I_1}}\Bigg|_{I_2=0}$$

It is called as i/p impedance (ohm)

$V_2 = Z_{21} I_1$

$$\boxed{Z_{21} = \frac{V_2}{I_1}}\Bigg|_{I_2=0}$$

It is called as forward transfer impedence

Put $I_1 = 0$ in ① & ②

$V_1 = Z_{12} I_2$

$$\boxed{Z_{12} = \frac{V_1}{I_2}}\Bigg|_{I_1=0}$$

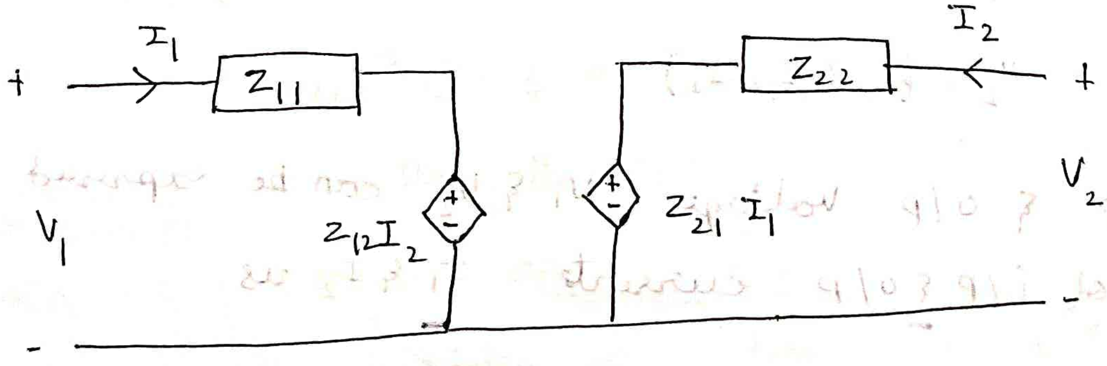
It is called reverse transfer impedence

$V_2 = Z_{22} I_2$

$$\boxed{Z_{22} = \frac{V_2}{I_2}}\Bigg|_{I_1=0}$$

It is called as o/p impedence (ohm)

- equivalent ckt: z parameter



→ <u>short ckt admittance (Y Parameter)</u>

$$I_1 = f_1(V_1, V_2)$$

$$I_2 = f_2(V_1, V_2)$$

$$I_1 = y_{11}V_1 + y_{12}V_2 \quad \ldots \quad (1)$$

$$I_2 = y_{21}V_1 + y_{22}V_2 \quad \ldots \quad (2)$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

Put $V_2 = 0$ in eq (1) & (2)

$I_1 = y_{11}V_1$

$\boxed{y_{11} = \frac{I_1}{V_1}}$ $\Big|_{V_2=0}$ Input admittance (mho)

$I_2 = y_{21}V_1$

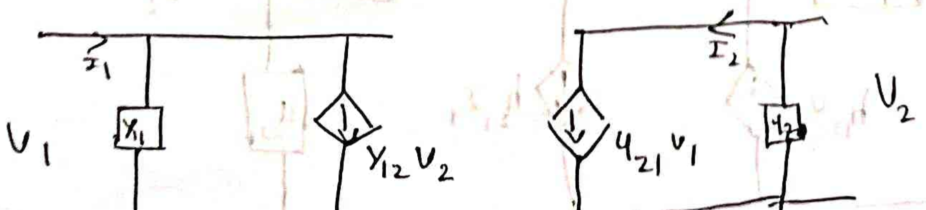
$\boxed{y_{21} = \frac{I_2}{V_1}}$ Forward transfer admittance

Put $V_1 = 0$ in eq (1) & (2)

$\boxed{y_{12} = \frac{I_1}{V_2}}$ $\Big|_{V_1=0}$ Reverse transfer admittance

$\boxed{y_{22} = \frac{I_2}{V_2}}$ $\Big|_{V_1=0}$ output admittance (mho)

• <u>equivalent y parameter model</u>



→ **H - Parameter :**

$$V_1 = f_1(I_1, V_2)$$

$$I_2 = f_2(I_1, V_2)$$

$$V_1 = H_{11} I_1 + H_{12} V_2 \quad \dots \quad (1)$$

$$I_2 = H_{21} I_1 + H_{22} V_2 \quad \dots \quad (2)$$

put $V_2 = 0$ in (1) & (2)

$$V_1 = H_{11} I_1$$

$$H_{11} = \frac{V_1}{I_1} \Big|_{V_2 = 0}$$ It is called as input impedance (ohm)

$$H_{21} = \frac{I_2}{I_1} \Big|_{V_2 = 0}$$ Forward current gain

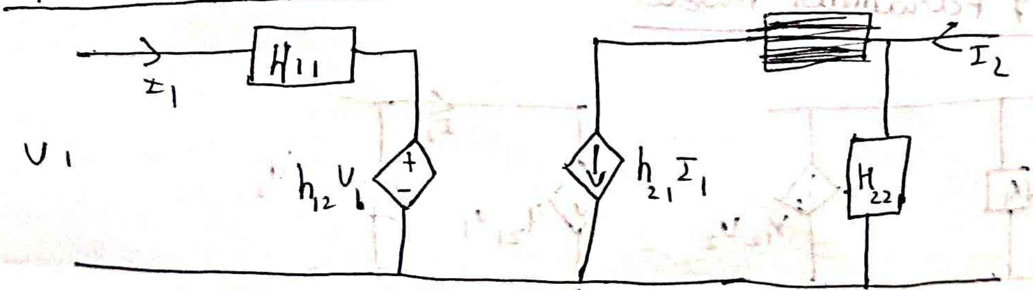
put $I_1 = 0$ in (1) & (2)

$$H_{12} = \frac{V_1}{V_2} \Big|_{I_1 = 0}$$ Reverse voltage gain

$$H_{22} = \frac{I_2}{V_2} \Big|_{I_1 = 0}$$ It is called as output admittance (mho)

→ It contains different parameters such as input impedance, current gain, AV & admittance hence called Hybrid parameter

→ **H parameter equivalent module:**



→ ABCD Parameters:

ABCD Parameter is used in Power transmission lines

Dependent: V_1 I_1

Independent: V_2 I_2

$$V_1 = f_1(V_2, I_2)$$

$$I_2 = f_1(V_2 I_2)$$

$$V_1 = A V_2 - B I_2 \quad \dots (1)$$

$$I_1 = C V_2 - D I_2 \quad \dots (2)$$

-ve sign for I_2 is due to the fact that I_2 flows in a direction opp^n to that of current

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

Put $I_2 = 0$ in (1) & (2)

$$A = \frac{V_1}{V_2} \Bigg|_{I_2 = 0} \qquad C = \cancel{I}\frac{I_1}{V_2} \Bigg|_{I_2 = 0}$$

Put $V_2 = 0$ in (1) & (2)

$$B = -\frac{V_1}{I_2} \Bigg|_{V_2 = 0} \qquad D = -\frac{I_1}{I_2} \Bigg|_{V_2 = 0}$$

show that $z = [y]^{-1}$

w know that, $[V] = [I][Z]$

Premultiply $[Z]^{-1}$ on both sides

$$[V][Z]^{-1} = [I]$$

$$[V][Z]^{-1} = [I]$$

$$[\cancel{V}][Z]^{-1} = [\cancel{V}][Y]$$

$$[Z]^{-1} = [Y]$$

$$[Y] = [Z]^{-1} = \frac{1}{[Z]}$$

-> Interrelation between parameters

Z —
- Y
- h
- ABCD

Y —
- Z
- h
- ABCD

H —
- Z
- Y
- ABCD

ABCD —
- Z
- Y
- H

• Z parameter in terms of Y parameter:

WKT, Z parameter are given by

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \ldots (1)$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \ldots (2)$$

y parameter are given by.

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \ldots (3)$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \ldots (4)$$

represent ③ & ④

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

V_1 & V_2 is obtained by cremers rule

$$V_1 = \frac{\begin{bmatrix} I_1 & Y_{12} \\ I_2 & Y_{22} \end{bmatrix}}{\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}} = \frac{Y_{22} I_1 - Y_{12} I_2}{Y_{11} Y_{22} - Y_{12} Y_{21}}$$

$$V_1 = \frac{Y_{22} I_1 - Y_{12} I_2}{\Delta Y}$$

$$V_1 = \frac{Y_{22} I_1}{\Delta Y} - \frac{Y_{12} I_2}{\Delta Y} \quad \ldots ⑤ \qquad Z_{11} = \frac{Y_{22}}{\Delta Y}, \quad Z_{12} = \frac{-Y_{12}}{\Delta Y}$$

comparing ⑤ & ①

$$\boxed{V_1 = \frac{Y_{22}}{\Delta Y} I_1 + \left(-\frac{Y_{12}}{\Delta Y}\right) I_2}$$

$$V_2 = \frac{\begin{bmatrix} Y_{11} & I_1 \\ Y_{21} & I_2 \end{bmatrix}}{Y_{11} Y_{22} - Y_{12} Y_{21}}$$

$$V_2 = \frac{Y_{11} I_2 - Y_{21} I_1}{\Delta Y}$$

$$V_2 = \frac{Y_{11}}{\Delta Y} I_2 - \frac{Y_{21}}{\Delta Y} I_1 \quad \ldots ⑥$$

Comparing ⑥ & ②

$$\boxed{V_2 = \frac{Y_{11}}{\Delta Y} I_2 + \left(-\frac{Y_{21}}{\Delta Y}\right) I_1} \qquad Z_{21} = \frac{Y_{11}}{\Delta Y}, \quad Z_{22} = -\frac{Y_{21}}{\Delta Y}$$

$$[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{Y_{22}}{\Delta Y} & -\frac{Y_{12}}{\Delta Y} \\ -\frac{Y_{21}}{\Delta Y} & \frac{Y_{11}}{\Delta Y} \end{bmatrix}$$

→ Z parameter in terms of H parameter.

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \ldots (1)$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \ldots (2)$$

H parameter

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \ldots (3)$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \ldots (4)$$

from (4), V_2

$$V_2 = \frac{I_2 - h_{21} I_1}{h_{22}}$$

$$V_2 = \frac{1}{h_{22}} I_2 - \frac{h_{21}}{h_{22}} I_1 \quad \ldots (5)$$

comparing (5) & (2)

$$\boxed{Z_{22} = \frac{1}{h_{22}}} \qquad \boxed{Z_{21} = \frac{-h_{21}}{h_{22}}}$$

subs V_2 in eqn (3)

$$V_1 = h_{11} I_1 + h_{12} \left[\frac{1}{h_{22}} I_2 - \frac{h_{21}}{h_{22}} I_1\right]$$

$$V_1 = h_{11} I_1 + \left[\frac{h_{12}}{h_{22}} I_2 - \frac{h_{21} h_{12}}{h_{22}} I_1\right]$$

$$V_1 = \frac{h_{12}}{h_{22}} I_2 + \left[\frac{h_{22} h_{11} - h_{21} h_{12}}{h_{22}}\right] I_1 \quad \ldots (6)$$

comparing (6) & (1)

$$\boxed{Z_{11} = \frac{\Delta h}{h_{22}}} \qquad \boxed{Z_{12} = \frac{h_{12}}{h_{22}}}$$

$$[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{\Delta h}{h_{22}} & \frac{h_{12}}{h_{22}} \\ -\frac{h_{21}}{h_{22}} & \frac{1}{h_{22}} \end{bmatrix}$$

-> Z parameter in terms of ABCD parameter.

Z parameter

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \ldots (1)$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \ldots (2)$$

ABCD parameter

$$V_1 = A V_2 - B I_2 \quad \ldots (3)$$

$$I_1 = C V_2 - D I_2 \quad \ldots (4)$$

from (4), V_2

$$\therefore V_2 = \frac{I_1 + D I_2}{C}$$

$$\boxed{V_2 = \frac{1}{C} I_1 + \frac{D}{C} I_2} \quad \ldots (5)$$

comparing (5) & (2)

$$\boxed{Z_{21} = \frac{1}{C}} \qquad \boxed{Z_{22} = \frac{D}{C}}$$

subs ⑤ in ③

$$V_1 = A\left[\frac{1}{C} I_1 + \frac{D}{C} I_2\right] - B I_2$$

$$V_1 = \frac{A}{C} I_1 + \frac{AD}{C} I_2 - B I_2$$

$$V_1 = \frac{A}{C} I_1 + \left[\frac{AD - BC}{C}\right] I_2$$

$$V_1 = \frac{A}{C} I_1 + \left[\frac{\Delta T}{C}\right] I_2 \quad \ldots ⑥$$

comparing ⑥ with ①

$$\boxed{Z_{11} = \frac{A}{C}} \qquad \boxed{Z_{12} = \frac{\Delta T}{C}}$$

$$[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{A}{C} & \frac{\Delta T}{C} \\ \frac{1}{C} & \frac{D}{C} \end{bmatrix}$$

Z - Parameter	Y parameter	h parameter	ABCD param
$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$	$\begin{bmatrix} \frac{Y_{22}}{\Delta Y} & \frac{-Y_{12}}{\Delta Y} \\ \frac{-Y_{21}}{\Delta Y} & \frac{Y_{11}}{\Delta Y} \end{bmatrix}$	$\begin{bmatrix} \frac{\Delta h}{h_{22}} & \frac{h_{12}}{h_{22}} \\ \frac{-h_{21}}{h_{22}} & \frac{1}{h_{22}} \end{bmatrix}$	$\begin{bmatrix} \frac{A}{C} & \frac{\Delta T}{C} \\ \frac{1}{C} & \frac{D}{C} \end{bmatrix}$

→ y in terms of z parameter:

y parameter

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \ldots (1)$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \ldots (2)$$

z parameter

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \ldots (3)$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \ldots (4)$$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$I_1 = \frac{\begin{vmatrix} V_1 & Z_{12} \\ V_2 & Z_{22} \end{vmatrix}}{\begin{vmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{vmatrix}} = \frac{V_1 Z_{22} - V_2 Z_{12}}{\Delta Z}$$

$$I_1 = \frac{Z_{22}}{\Delta Z} V_1 - \frac{Z_{12}}{\Delta Z} V_2 \quad \ldots (5)$$

comparing (5) & (1)

$$Y_{11} = \frac{Z_{22}}{\Delta Z} \qquad Y_{12} = \frac{-Z_{12}}{\Delta Z}$$

$$I_2 = \frac{\begin{vmatrix} Z_{11} & V_1 \\ Z_{21} & V_2 \end{vmatrix}}{\begin{vmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{vmatrix}} = \frac{Z_{11} V_2 - Z_{21} V_1}{\Delta Z}$$

$$I_2 = \frac{Z_{11}}{\Delta Z} V_2 - \frac{Z_{21}}{\Delta Z} V_1 \quad \ldots (6)$$

comparing (6) & (2)

$$Y_{21} = \frac{-Z_{21}}{\Delta Z} \qquad Y_{22} = \frac{Z_{11}}{\Delta Z}$$

$$[Y] = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} \frac{Z_{22}}{\Delta Z} & -\frac{Z_{12}}{\Delta Z} \\ \frac{-Z_{21}}{\Delta Z} & \frac{Z_{11}}{\Delta Z} \end{bmatrix}$$

-> **y parameters in terms of h parameters :**

y parameters :

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \ldots (1)$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \ldots (2)$$

h parameters :

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \ldots (3)$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \ldots (4)$$

from eqn (3),

$$I_1 = \frac{V_1 - h_{12} V_2}{h_{11}}$$

$$I_1 = \frac{1}{h_{11}} V_1 + \left(-\frac{h_{12}}{h_{11}} \right) V_2$$

$$I_1 = \frac{1}{h_{11}} V_1 - \frac{h_{12}}{h_{11}} V_2 \quad \ldots (5)$$

comparing (5) & (1)

$$\boxed{Y_{11} = \frac{1}{h_{11}}} \qquad \boxed{Y_{12} = \frac{-h_{12}}{h_{11}}}$$

subs (5) in (4)

$$I_2 = h_{21} \left(\frac{1}{h_{11}} V_1 - \frac{h_{12}}{h_{11}} V_2 \right) + h_{22} V_2$$

$$I_2 = \frac{h_{21}}{h_{11}} V_1 + \frac{h_{21} h_{12} - h_{11} h_{22}}{h_{11}} V_2 \quad \ldots (6)$$

comparing 6 & 2

$$\boxed{y_{21} = \frac{h_{21}}{h_{11}}} \qquad \boxed{y_{22} = \frac{\Delta h}{h_{11}}}$$

$$\therefore [Y] = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} \frac{1}{h_{11}} & -\frac{h_{12}}{h_{11}} \\ \frac{h_{21}}{h_{11}} & \frac{\Delta h}{h_{11}} \end{bmatrix}$$

→ y parameters in terms of abcd parameters:

y parameters :

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \ldots (1)$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \ldots (2)$$

abcd parameters :

$$V_1 = A V_2 - \cancel{B V_2}\ B I_2 \quad \ldots (3)$$

$$I_1 = C V_2 - \cancel{D V_2}\ D I_2 \quad \ldots (4)$$

from eqn 3

$$I_2 = -\frac{1}{B} V_1 + \frac{A}{B} V_2 \quad \ldots (5)$$

comparing 5 & 2

$$\boxed{Y_{21} = -\frac{1}{B}} \qquad \boxed{Y_{22} = \frac{A}{B}}$$

subs 5 in 4

$$I_1 = C V_2 - D\left(\frac{A}{B} V_2 - \frac{1}{B} V_1\right)$$

$$I_1 = C V_2 - \frac{DA}{B} V_2 + \frac{D}{B} V_1$$

$$I_1 = \frac{D}{B} V_1 + \frac{C - DA}{B} V_2$$

$$I_1 = \frac{D}{B} V_1 + \frac{BC - DA}{B} V_2$$ $\rightarrow -(AD-$

since $BC - DA$ this is $\boxed{-(DA - BC)}$ by rearranging

$$I_1 = \frac{D}{B} V_1 - \frac{\Delta T}{B} V_2 \quad \ldots (6)$$

comparing eqn (6) & (1)

$$\boxed{Y_{11} = \frac{D}{B}} \qquad \boxed{Y_{12} = \frac{-\Delta T}{B}}$$

$$\therefore [Y] = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} = \begin{bmatrix} \frac{D}{B} & -\frac{\Delta T}{B} \\ -\frac{1}{B} & \frac{A}{B} \end{bmatrix}$$

y - parameter	z - parameter	h - parameter	ABCD parameter
$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}$	$\begin{bmatrix} \frac{z_{22}}{\Delta z} & -\frac{z_{12}}{\Delta z} \\ -\frac{z_{21}}{\Delta z} & \frac{z_{11}}{\Delta z} \end{bmatrix}$	$\begin{bmatrix} \frac{1}{h_{11}} & -\frac{h_{12}}{h_{11}} \\ \frac{h_{21}}{h_{11}} & \frac{\Delta h}{h_{11}} \end{bmatrix}$	$\begin{bmatrix} \frac{D}{B} & -\frac{\Delta T}{B} \\ -\frac{1}{B} & \frac{A}{B} \end{bmatrix}$

→ h parameter in terms of z parameter

h parameter

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \ldots (1)$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \ldots (2)$$

z parameter

$$V_1 = z_{11} I_1 + z_{12} I_2 \quad \ldots (3)$$

$$V_2 = z_{21} I_1 + z_{22} I_2 \quad \ldots (4)$$

from eqn ④

$$I_2 = \frac{V_2 - Z_{21} I_1}{Z_{22}}$$

$$I_2 = \frac{1}{Z_{22}} V_2 - \frac{Z_{21}}{Z_{22}} I_1 \quad \cdots ⑤$$

comparing ⑤ and ②

$$\boxed{h_{22} = \frac{1}{Z_{22}}} \qquad \boxed{h_{21} = \frac{-Z_{21}}{Z_{22}}}$$

subs eqn ⑤ in ③

$$V_1 = Z_{11} I_1 + Z_{12} \left(\frac{1}{Z_{22}} V_2 - \frac{Z_{21}}{Z_{22}} I_1 \right)$$

$$V_1 = Z_{11} I_1 + \frac{Z_{12}}{Z_{22}} V_2 - \frac{Z_{12} Z_{21}}{Z_{22}} I_1$$

$$V_1 = \frac{Z_{12}}{Z_{22}} V_2 + \frac{Z_{11} Z_{22} - Z_{12} Z_{21}}{Z_{22}} I_1$$

$$V_1 = \frac{Z_{12}}{Z_{22}} V_2 + \frac{\Delta Z}{Z_{22}} I_1 \quad \cdots ⑥$$

comparing ⑥ & ①

$$\boxed{h_{11} = \frac{\Delta Z}{Z_{22}}} \qquad \boxed{h_{12} = \frac{Z_{12}}{Z_{22}}}$$

$$\therefore [h] = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{\Delta Z}{Z_{22}} & \frac{Z_{12}}{Z_{22}} \\ \frac{-Z_{21}}{Z_{22}} & \frac{1}{Z_{22}} \end{bmatrix}$$

→ h parameter in terms of Y parameter.

h parameter :

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \cdots (1)$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \cdots (2)$$

Y parameter :

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \cdots (3)$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \cdots (4)$$

from eqn (3)

$$V_1 = \frac{I_1}{Y_{11}} - \frac{Y_{12}}{Y_{11}} V_2$$

$$V_1 = \frac{1}{Y_{11}} I_1 - \frac{Y_{12}}{Y_{11}} V_2 \quad \cdots (5)$$

comparing (5) & (1)

$$\boxed{h_{11} = \frac{1}{Y_{11}}} \qquad \boxed{h_{12} = \frac{-Y_{12}}{Y_{11}}}$$

subs eqn (5) in (4)

$$I_2 = Y_{21}\left(\frac{1}{Y_{11}} I_1 - \frac{Y_{12}}{Y_{11}} V_2\right) + Y_{22} V_2$$

$$I_2 = \frac{Y_{21}}{Y_{11}} I_1 - \frac{Y_{21} Y_{12}}{Y_{11}} V_2 + Y_{22} V_2$$

$$I_2 = \frac{Y_{21}}{Y_{11}} I_1 + \frac{Y_{22} Y_{11} - Y_{21} Y_{12}}{Y_{11}} V_2 \quad \cdots (6)$$

comparing ⑥ & ②

$$\boxed{h_{21} = \frac{y_{21}}{y_{11}}} \qquad \boxed{h_{22} = \frac{\Delta y}{y_{11}}}$$

$$[h] = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{1}{y_{11}} & -\frac{y_{12}}{y_{11}} \\ \frac{y_{21}}{y_{11}} & \frac{\Delta y}{y_{11}} \end{bmatrix}$$

→ h parameter in terms of abcd parameter:

h parameter:

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \ldots (1)$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \ldots (2)$$

abcd parameter:

$$V_1 = A V_2 - B I_2 \quad \ldots (3)$$

$$I_1 = C V_2 - D I_2 \quad \ldots (4)$$

from eqn ④

$$I_2 = \frac{C V_2 - I_1}{D}$$

$$I_2 = \frac{C}{D} V_2 - \frac{1}{D} I_1 \quad \ldots (5)$$

comparing ⑤ & ②

$$\boxed{h_{21} = \frac{-1}{D}} \qquad \boxed{h_{22} = \frac{C}{D}}$$

subs ⑤ in ③

$$V_1 = A V_2 - B\left(\frac{C}{D} V_2 - \frac{1}{D} I_1\right)$$

$$V_1 = AV_2 - B\left(\frac{C}{D}V_2 - \frac{1}{D}I_1\right)$$

$$V_1 = \frac{B}{D}I_1 + A - \frac{BC}{D}V_2$$

$$V_1 = \frac{B}{D}I_1 + \frac{DA - BC}{D}V_2$$

$$V_1 = \frac{B}{D}I_1 + \frac{\Delta T}{D}V_2 \quad \ldots (6)$$

comparing (6) & (1)

$$\boxed{h_{11} = \frac{B}{D}} \qquad \boxed{h_{12} = \frac{\Delta T}{D}}$$

$$\therefore [h] = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{B}{D} & \frac{\Delta T}{D} \\ \frac{-1}{D} & \frac{C}{D} \end{bmatrix}$$

h Parameter	Z parameter	Y parameter	abcd parameter
$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}$	$\begin{bmatrix} \frac{\Delta Z}{Z_{22}} & \frac{Z_{12}}{Z_{22}} \\ \frac{-Z_{21}}{Z_{22}} & \frac{1}{Z_{22}} \end{bmatrix}$	$\begin{bmatrix} \frac{1}{Y_{11}} & \frac{-Y_{12}}{Y_{11}} \\ \frac{Y_{21}}{Y_{11}} & \frac{\Delta Y}{Y_{11}} \end{bmatrix}$	$\begin{bmatrix} \frac{B}{D} & \frac{\Delta T}{D} \\ \frac{-1}{D} & \frac{C}{D} \end{bmatrix}$

-> ABCD Parameter in terms of z parameter:

ABCD parameter

$$V_1 = AV_2 - BI_2 \quad \ldots (1)$$

$$I_1 = CV_2 - DI_2 \quad \ldots (2)$$

Z parameter

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \quad \ldots (3)$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2 \quad \ldots (4)$$

from 4

$$I_1 = \frac{V_2 - Z_{22} I_2}{Z_{21}} \quad \ldots$$

$$I_1 = \frac{-Z_{22}}{Z_{21}} I_2 + \frac{1}{Z_{21}} V_2 \quad \ldots 5$$

~~subs~~ 5 & 2

$$\boxed{D = \frac{Z_{22}}{Z_{21}}} \qquad \boxed{C = \frac{1}{Z_{21}}}$$

subs I_1 in 3

$$V_1 = Z_{11}\left(\frac{V_2}{Z_{21}} - \frac{Z_{22}}{Z_{21}} I_2\right) - Z_{12} I_2$$

$$V_1 = \frac{Z_{11}}{Z_{21}} V_2 - \frac{(Z_{11}Z_{22} - Z_{12}Z_{21})}{Z_{21}} I_2 \quad \ldots 6$$

comparing eqn 6 and 1

$$\boxed{A = \frac{Z_{11}}{Z_{21}}} \qquad \boxed{B = \frac{\Delta Z}{Z_{21}}}$$

$$\therefore \begin{bmatrix} ABCD \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{Z_{11}}{Z_{21}} & \frac{\Delta Z}{Z_{21}} \\ \frac{1}{Z_{21}} & \frac{Z_{22}}{Z_{21}} \end{bmatrix}$$

-> ABCD Parameters in terms of Y Parameter :

ABCD parameter :

$$V_1 = A V_2 - B I_2 \quad \ldots 1$$

$$I_1 = C V_2 - D I_2 \quad \ldots 2$$

~~Z~~ Y Parameter :

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \ldots 3$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \ldots 4$$

from ④

$$V_1 = \frac{I_2 - Y_{22} V_2}{Y_{21}}$$

$$V_1 = \frac{1}{Y_{21}} I_2 - \frac{Y_{22}}{Y_{21}} V_2 \quad \ldots ⑤$$

comparing ⑤ and ①

$$\boxed{A = -\frac{Y_{22}}{Y_{21}}} \qquad \boxed{B = -\frac{1}{Y_{21}}}$$

subs ⑤ in ③

$$I_1 = \frac{Y_{11}}{Y_{21}} I_2 - \frac{Y_{11} Y_{22}}{Y_{21}} V_2 + Y_{12} V_2$$

$$I_1 = \frac{Y_{11}}{Y_{21}} I_2 + \left(\frac{- Y_{11} Y_{22} + Y_{12} Y_{21}}{Y_{21}} V_2\right) \quad \ldots ⑥$$

comparing ⑥ & ②

$$\boxed{D = -\frac{Y_{11}}{Y_{21}}} \qquad \boxed{C = -\frac{\Delta Y}{Y_{21}}}$$

$$\therefore \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} -\frac{Y_{22}}{Y_{21}} & -\frac{1}{Y_{21}} \\ -\frac{\Delta Y}{Y_{21}} & -\frac{Y_{11}}{Y_{21}} \end{bmatrix}$$

→ ABCD in terms of h parameters

ABCD Parameters:

$$V_1 = A V_2 - B I_2 \quad \ldots ①$$

$$I_1 = C V_2 - D I_2 \quad \ldots ②$$

h parameters:

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \ldots (3)$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \ldots (4)$$

from 4,

$$I_1 = \frac{I_2}{h_{21}} - \frac{h_{22} V_2}{h_{21}} \quad \ldots (5)$$

comparing (5) & (2)

$$\boxed{C = \frac{-h_{22}}{h_{21}}} \qquad \boxed{D = -\frac{1}{h_{21}}}$$

subs (5) in (3)

$$V_1 = \frac{h_{11}}{h_{21}} I_2 - \frac{h_{22} h_{11}}{h_{21}} V_2 + h_{12} V_2$$

$$V_1 = \frac{h_{11}}{h_{21}} I_2 + \left(\frac{-h_{22} h_{11} + h_{12} h_{21}}{h_{21}} \right) V_2 \quad \ldots (6)$$

comparing (6) & (1)

$$\boxed{A = -\frac{\Delta h}{h_{21}}} \qquad \boxed{B = \frac{-h_{11}}{h_{21}}}$$

$$\therefore \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{\Delta h}{h_{21}} & \frac{-h_{11}}{h_{21}} \\ \frac{-h_{22}}{h_{21}} & -\frac{1}{h_{21}} \end{bmatrix}$$

	z parameters	y parameters	h parameters
$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$	$\begin{bmatrix} \frac{z_{11}}{z_{21}} & \frac{\Delta z}{z_{21}} \\ \frac{1}{z_{21}} & \frac{z_{22}}{z_{21}} \end{bmatrix}$	$\begin{bmatrix} -\frac{y_{22}}{y_{21}} & -\frac{1}{y_{21}} \\ -\frac{\Delta y}{y_{21}} & -\frac{y_{11}}{y_{21}} \end{bmatrix}$	$\begin{bmatrix} -\frac{\Delta h}{h_{21}} & -\frac{h_{11}}{h_{21}} \\ -\frac{h_{22}}{h_{21}} & -\frac{1}{h_{21}} \end{bmatrix}$

→ Problems :

1) Find Y & Z parameter for the circuit shown

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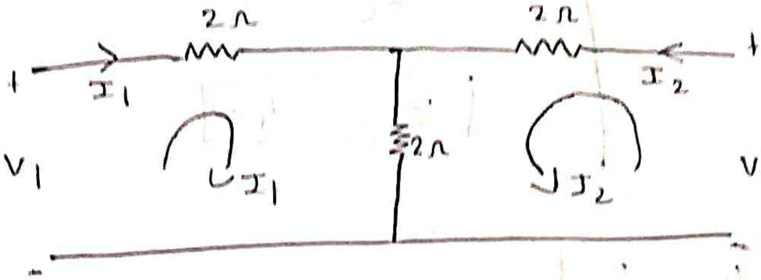


soln: Z parameter are given by :

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

Y parameter are given by :

$$I_1 = Y_{11} V_1 + Y_{12} V_2$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2$$

apply KVL to loop 1

$$-2 I_1 - 2(I_1 + I_2) + V_1 = 0$$

$$-V_1 = -4 I_1 - 2 I_2 \quad \text{or} \quad V_1 = 4 I_1 + 2 I_2 \ldots (1)$$

apply KVL to loop 2

$$-2 I_2 - 2 (I_2 + I_1) + V_2 = 0$$

$$V_2 = 4 I_2 + 2 I_1 \ldots (2)$$

subs $I2 = 0$ in eqn (1) & (2)

$$\frac{V_1}{I_1} = 4 = Z_{11}$$

$$\boxed{\therefore Z_{11} = 4\Omega}$$

$$\frac{V_2}{I_1} = 2 = Z_{21} \qquad \boxed{\therefore Z_{21} = 2\Omega}$$

Put $I_1 = 0$ in 1 & 2

$$\frac{V_1}{I_2} = 2$$

$$\boxed{Z_{21} = 2\,\Omega}$$

$$\frac{V_2}{I_2} = 4$$

$$\boxed{Z_{22} = 4\,\Omega}$$

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$$

Put $V_2 = 0$ in eqn 2

$$0 = 4I_2 + 2I_1$$

$$-4I_2 = 2I_1$$

$$\boxed{I_2 = -\frac{1}{2} I_1}$$

subs I_2 in 1

$$V_1 = 4I_1 + 2\left(-\frac{1}{2} I_1\right)$$

$$V_1 = 4I_1 - I_1$$

$$V_1 = 3I_1$$

$$\frac{V_1}{I_1} = 3 \quad = \frac{I_1}{V_1} = \frac{1}{3}\,\mho$$

$$\boxed{Y_{11} = \frac{1}{3}\,\mho}$$

$$\boxed{I_1 = \frac{V_1}{3}}$$

$$I_2 = -\frac{1}{2}\left(\frac{V_1}{3}\right)$$

$$\boxed{\frac{I_2}{V_1} = -\frac{1}{6}\,\mho}$$

put $V_1 = 0$ in eqn ①

$$0 = 4 I_1 + 2 I_2$$

$$-4 I_1 = 2 I_2$$

$$\boxed{I_2 = -2 I_1}$$

sub I_2 in

$$V_2 = 2 I_1 + 4(-2 I_1)$$

$$V_2 = 2 I_1 - 8 I_1$$

$$V_2 = -6 I_1$$

$$\frac{V_2}{I_2} = -6 \qquad \frac{I_2}{V_2} = -\frac{1}{6} = Y_{12}$$

$$\boxed{Y_{12} = -\frac{1}{6}}$$

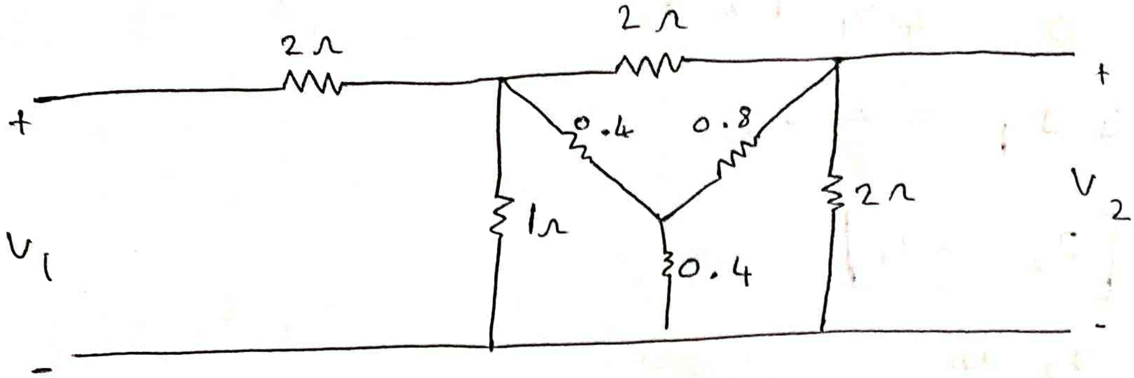
$$I_1 = \frac{V_2}{-6}$$

$$I_2 = \frac{\cancel{2}^{1} V_2}{\cancel{6}_{3}}$$

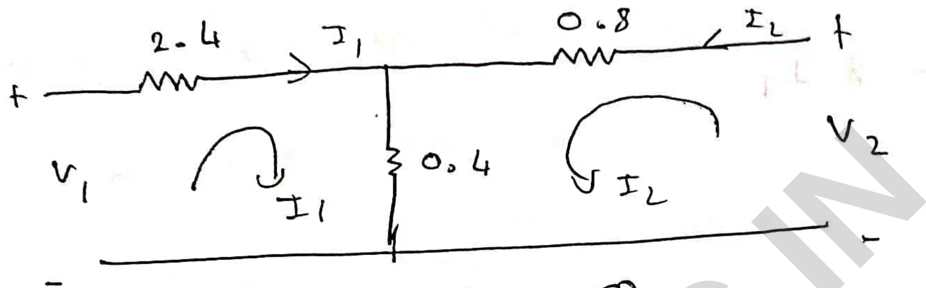
$$\boxed{\frac{I_2}{V_2} = \frac{1}{3}}$$

$$[Y] = \begin{bmatrix} \frac{1}{3} & -\frac{1}{6} \\ -\frac{1}{6} & \frac{1}{3} \end{bmatrix}$$

2) Find Z, Y, ABCD parameter for the ckt shown.



soln:



apply KVL to loop ①

$-2.4 I_1 - 0.4 I_1 - 0.4 I_2 + V_1 = 0$

$V_1 = 2.8 I_1 + 0.4 I_2$ ... ①

apply KVL to loop ②

$-0.8 I_2 - 0.4 I_2 - 0.4 I_1 + V_2 = 0$

$V_2 = 0.4 I_1 + 1.2 I_2$ ... ②

Z parameters

$V_1 = z_{11} I_1 + z_{12} I_2$

$V_2 = z_{21} I_1 + z_{22} I_2$

put $I_2 = 0$ in ① & ②

$\frac{V_1}{I_1} = 2.8$ $\boxed{z_{11} = 2.8\,\Omega}$

$\frac{V_2}{...} = z_{21} = 0.4 = \boxed{z_{21} = 0.4}$

put $I_1 = 0$ in ① & ②

$\frac{V_1}{I_2} = 0.4$ $\boxed{Z_{12} = 0.4}$

$\frac{V_2}{I_2} = 1.2$ $\boxed{Z_{22} = 1.2}$

$$[Z] = \begin{bmatrix} 2.8 & 0.4 \\ 0.4 & 1.2 \end{bmatrix}$$

$$\Delta Z = 2.8(1.2) - (0.4)(0.4)$$

$$\boxed{\Delta Z = 3.2}$$

$$\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} = \begin{bmatrix} \frac{Z_{22}}{\Delta Z} & -\frac{Z_{12}}{\Delta Z} \\ -\frac{Z_{21}}{\Delta Z} & \frac{Z_{11}}{\Delta Z} \end{bmatrix}$$

$$[y] = \begin{bmatrix} \frac{1.2}{3.2} & -\frac{0.4}{3.2} \\ -\frac{0.4}{3.2} & \frac{2.8}{3.2} \end{bmatrix}$$

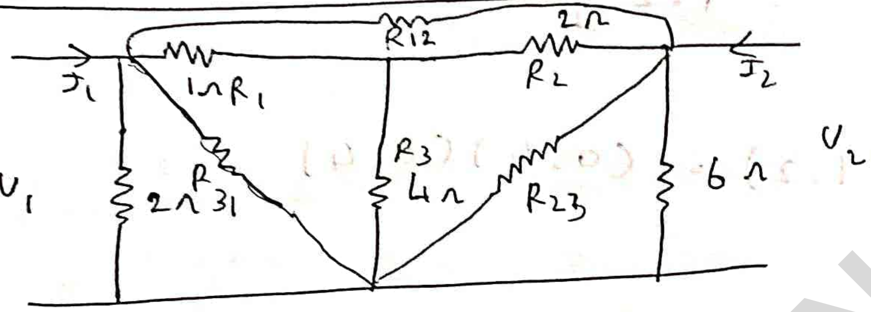
$$[y] = \begin{bmatrix} 0.375 & -0.125 \\ -0.125 & 0.875 \end{bmatrix}$$

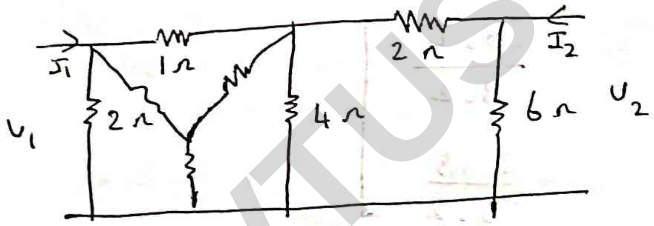
$$[ABCD] = \begin{bmatrix} \frac{Z_{11}}{Z_{21}} & \frac{\Delta Z}{Z_{21}} \\ \frac{1}{Z_{21}} & \frac{Z_{22}}{Z_{21}} \end{bmatrix} = \begin{bmatrix} \frac{2.8}{0.4} & \frac{3.2}{0.4} \\ \frac{1}{0.4} & \frac{1.2}{0.4} \end{bmatrix}$$

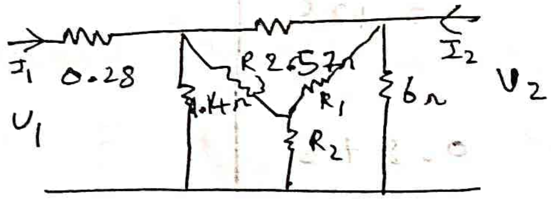
$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 7 & 8 \\ 2.5 & 3 \end{bmatrix}$$

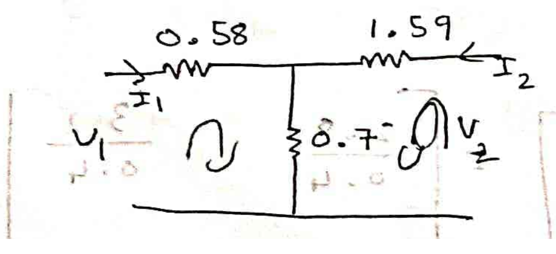
$\therefore \; A = 7, \quad B = 8, \quad C = 2.5, \quad D = 3$

3)









Z parameters

$V_1 = Z_{11} I_1 + Z_{12} I_2$

$V_2 = Z_{21} I_1 + Z_{22} I_2$

apply KVL to loop ①

$-0.58 I_1 - 0.7 I_1 - 0.7 I_2 + V_1 = 0$

$-1.28 I_1 - 0.7 I_2 + V_1 = 0$

$\boxed{V_1 = 1.28 I_1 + 0.7 I_2} \quad \ldots ①$

apply KVL to loop ②

$-1.59 I_2 - 0.7 I_2 - 0.7 I_1 + V_2 = 0$

$\boxed{V_2 = 0.7 I_1 + 2.29 I_2} \quad \ldots ②$

put $I_2 = 0$ in ① & ②

$\frac{V_1}{I_1} = 1.28 \Rightarrow \boxed{Z_{11} = 1.28}$

$\frac{V_2}{I_1} = 0.7 \Rightarrow \boxed{Z_{21} = 0.7}$

put $I_1 = 0$ in ① & ②

$\frac{V_1}{I_2} = 0.7 \Rightarrow \boxed{Z_{12} = 0.7}$

$\frac{V_2}{I_2} = 2.29 \Rightarrow \boxed{Z_{22} = 2.29}$

$$\therefore [Z] = \begin{bmatrix} 1.28 & 0.7 \\ 0.7 & 2.29 \end{bmatrix}$$

Y parameters:

$I_1 = Y_{11} V_1 + Y_{12} V_2$

$I_2 = Y_{21} V_1 + Y_{22} V_2$

Put $V_2 = 0$ in eqn ②

$0 = 0.7 I_1 + 2.29 I_2$

$I_2 = \frac{-0.7}{2.29} I_1$

$\boxed{I_2 = -0.30 I_1}$

subs I_2 in ①

$V_1 = 1.28\ I_1 + 0.7\ I_2$

$V_1 = 1.28\ I_1 + 0.7(-0.3)\ I_1$

$\frac{V_1}{I_1} = 1.07 \Rightarrow \frac{I_1}{V_1} = 0.93$

$\boxed{Y_{11} = 0.93}$

$I_1 = \frac{V_1}{0.\cancel{4}93}$

~~$I_2 =$~~ $I_2 = 0.305 \left(\frac{V_1}{0.\cancel{4}93}\right)$

$\frac{I_2}{V_1} = \cancel{\frac{0.305}{0.493}} = \frac{-0.30}{1.070} = -0.28\cancel{4}$

$\boxed{Y_{21} = -0.28\ \mho}$

subs $V_1 = 0$ in eqn ① & ②

$0 = 1.28\ I_1 + 0.70\ I_2$

$I_2 = -\frac{1.28}{0.70}\ I_1$

$I_2 = -1.82\ I_1$

subs I_2 in ②

$V_2 = 0.70\ I_1 + 2.28(-1.82)\ I_1$

$V_2 = 0.70\ I_1 - 4.14\ I_1$

$V_2 = -3.44\ I_1$

$\boxed{= -0.29\ \mho = Y_{12}}$

$V_2 = -3.44\ I_1$

$I_2 = \frac{-1.52}{-3.44}\ V_2$

$\boxed{\frac{I_2}{V_2} = 0.52\ \mho}$

$$[Y] = \begin{bmatrix} 0.93 & -0.29 \\ -0.28 & 0.52 \end{bmatrix}$$

for h parameter :

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \ldots (1)$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \ldots (2)$$

$$V_1 = 1.28\, I_1 + 0.7\, I_2 \quad \ldots (3)$$

$$V_2 = 0.7\, I_1 + 2.29\, I_2 \quad \ldots (4)$$

$$I_2 = \frac{V_2 - 0.7\, I_1}{2.29}$$

$$I_2 = 0.43\, V_2 - 0.3056\, I_1 \quad \ldots (5)$$

comparing (5) with (2)

$$\boxed{h_{21} = -0.30} \qquad \boxed{h_{22} = 0.43}$$

sub I_2 in V_1

$$V_1 = 1.28\, I_1 + 0.7 \left(\frac{V_2 - 0.7\, I_1}{2.29} \right)$$

$$V_1 = 1.28\, I_1 + \frac{0.7\, V_2}{2.29} - 0.30\, I_1$$

$$V_1 = 0.30\, V_2 + 1.08\, I_1 \quad \ldots (6)$$

comparing (6) & (1)

$$\boxed{h_{11} = 1.07} \qquad \boxed{h_{12} = 0.30}$$

$$[h] = \begin{bmatrix} \cancel{0.98}\ 1.06 & 0.30 \\ -0.30 & 0.43 \end{bmatrix}$$

ABCD

$V_1 = 1.28 I_1 + 0.70 I_2$ ... (1)

$V_2 = 0.70 I_1 + 2.28 I_2$ ... (2)

$V_1 = A V_2 - B I_2$ ... (3)

$I_1 = C V_2 - D I_2$ ... (4)

from (2)

$$I_1 = \frac{V_2 - 2.28 I_2}{0.7} \quad \text{... (5)}$$

comparing (5) & (4)

$\boxed{I_1 = C = 1.42}$ $\boxed{D = 3.25}$

~~Sub~~ subs I_1 in V_1

$$V_1 = 1.28 (1.42 V_2 - 3.25 I_2) + 0.70 I_2$$

$\boxed{V_1 = 1.81 V_2 - 3.45 I_2}$

$\boxed{A = 1.83}$ $\boxed{B = 3.45}$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1.83 & 3.45 \\ 1.42 & 3.25 \end{bmatrix}$$

Note :

1) For a Reciprocal network

$$z_{12} = z_{21}$$

$$Y_{12} = Y_{21}$$

$$h_{12} = -h_{21}$$

2) For a symetrical network

$$z_{11} = z_{22}$$

$$Y_{11} = Y_{22}$$

$$A = D$$

- Applications of h, Z, T are

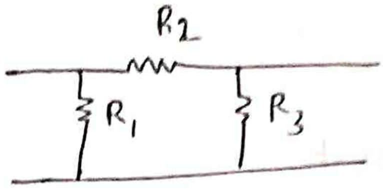
h parameter : It is used in analysis of small signal amplifier using BJT. The equivalent circuit of hparameter can be used as an AC equivalent ckt of a small signal BJT amplifier To carry out AC analysis of amplifier

Z parameter : It is used to represent given two port network into equivalent T n/w which is the most important n/w in communication n/w

T parameter : They are generally used for analysis of power transmission line where I/P port is a sending end and O/P port of the n/w is the receiving end of the line

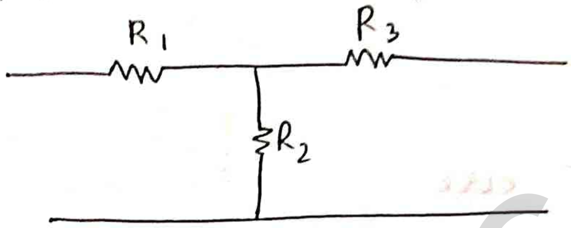
- If there is no dependent source in the given n/w then $Z_{12} = Z_{21}$ & $Y_{12} = Y_{21}$ if a n/w is said to be reciprocal

- If the n/w is of pie type

π 

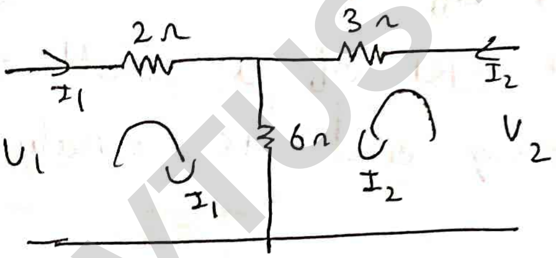
Then apply KCL

- If the n/w is of T Type



The apply KVL

→ find z and y parameter for n/w shown:



z parameter

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \ldots (1)$$

$$V_2 = Z_{21} I_1 = Z_{22} I_2 \quad \ldots (2)$$

apply KVL to loop (1)

$$-2 I_1 - 6(I_1 + I_2) + V_1 = 0$$

$$V_1 = 8 I_1 + 6 I_2 \quad \ldots (3)$$

apply KVL to loop (2)

$$-3I_2 - 6(I_2 + I_1) + V_2 = 0$$

$$V_2 = 6I_1 + 9I_2 \quad \ldots (4)$$

Put $I_2 = 0$ in (3) & (4)

$$\frac{V_1}{I_1} = 8 \Rightarrow \boxed{Z_{11} = 8\,\Omega} \qquad \frac{V_2}{I_2} = 6 \Rightarrow \boxed{Z_{21} = 6}$$

Put $I_1 = 0$ in (3) & (4)

$$\frac{V_2}{I_2} = 9 \quad \boxed{Z_{22} = 9} \qquad \frac{V_1}{I_2} = 6 \quad \boxed{Z_{12} = 6}$$

$$[Z] = \begin{bmatrix} 8 & 6 \\ 6 & 9 \end{bmatrix}$$

Put $V_2 = 0$ in (4)

$$-6I_1 \pm 9I_2$$

$$I_2 = -\frac{6}{9} I_1$$

$$I_2 = -0.66 I_1$$

subs I_2 in (3)

$$V_1 = 8I_1 + 6(-0.66)I1$$

$$\frac{V_1}{I_1} = 4.04 \qquad \frac{I_1}{V_1} = \frac{1}{4.04} = 0.25 = Y_{11}$$

$$\boxed{\Delta Z = 36}$$

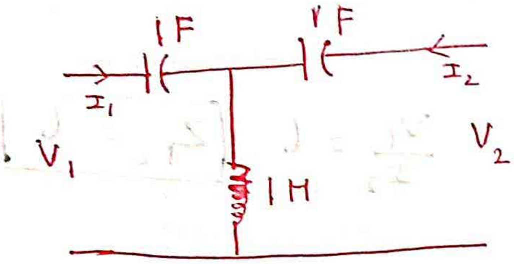
$$Y_{12} = -\frac{Z_{12}}{\Delta Z} = -\frac{6}{36} = -0.16\,\mho$$

$$Y_{21} = -\frac{Z_{21}}{\Delta Z} = -\frac{6}{36} = -0.16\,\mho$$

$$Y_{22} = \frac{Z_{11}}{\Delta Z} = \frac{8}{36} = 0.22\,\mho$$

$$[Y] = \begin{bmatrix} 0.25 & -0.16 \\ -0.16 & 0.22 \end{bmatrix}$$

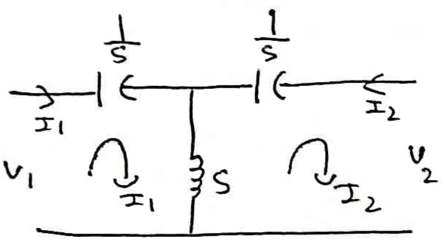
→ Find Z, Y, h, ABCD



laplace transform of 1F is $\frac{1}{S}$

1H is S

soln:



Z parameters

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \cdots ①$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \cdots ②$$

apply KVL to loop ①

$$V_1 - \frac{1}{S} I_1 - S(I_1 + I_2) = 0$$

$$V_1 - \frac{1}{S} I_1 - S I_1 - S I_2 = 0$$

$$V_1 = \left(\frac{1 + S^2}{S}\right) I_1 + S I_2 \quad \cdots ③$$

apply KVL to loop 2

$$-\frac{1}{S} I_2 - S I_2 - S I_1 + V_2 = 0$$

$$V_2 = S I_1 + \left(\frac{1 + S^2}{S}\right) I_2 \quad \cdots ④$$

put $I_2 = 0$ in ③ & ④

$$\frac{V_1}{I_1} = \boxed{z_{11} = \frac{1+s^2}{s}} \qquad \frac{V_2}{I_1} = s \qquad \boxed{z_{21} = s}$$

put $I_1 = 0$ in ③ & ④

$$\frac{V_1}{I_2} = \boxed{z_{12} = s} \qquad \frac{V_2}{I_2} = \frac{1+s^2}{s} \qquad \boxed{z_{22} = \frac{1+s^2}{s}}$$

$$Z = \begin{bmatrix} \frac{1+s^2}{s} & s \\ s & \frac{1+s^2}{s} \end{bmatrix}$$

---

y parameters are given by

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \ldots ①$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \ldots ②$$

$$V_1 = \left(\frac{1+s^2}{s}\right) I_1 + s I_2 \quad \ldots ③$$

$$V_2 = s I_1 + \left(\frac{1+s^2}{s}\right) I_2 \quad \ldots ④$$

put $V_2 = 0$ in eqn ④

$$0 = s I_1 + \left(\frac{1+s^2}{s}\right) I_2$$

$$s I_1 = -\frac{1+s^2}{s} I_2$$

$$I_1 = -\left(\frac{1+s^2}{s^2}\right) I_2$$

$$I_2 = \frac{-s^2}{1+s^2} I_1$$

$$V_1 = \left(\frac{1+s^2}{s}\right) \cdot -\left(\frac{1+s^2}{s^2}\right) I_2 + s I_2$$

$$V_1 = -\frac{(1+s^2)^2}{s^3} I_2 + s I_2$$

$$V_1 = \frac{s^4 - (1+s^2)^2}{s^3} I_2$$

$$V_1 = \frac{s^4 - (1+s^4+2s^2)}{s^3} I_2$$

$$V_1 = \frac{\cancel{s^4} - 1 - \cancel{s^4} - 2s^2}{s^3} I_2$$

$$\frac{I_2}{V_1} = \frac{-s^3}{(1+2s^2)}$$

$$\boxed{Y_{21} = \frac{-s^3}{1+2s^2}}$$

subs I_2 in eqn 3

$$V_1 = \left(\frac{1+s^2}{s}\right) I_1 + s\left(\frac{-s^2}{1+s^2}\right) I_1$$

$$V_1 = \left(\frac{1+s^2}{s} - \frac{s^3}{1+s^2}\right) I_1$$

$$V_1 = \frac{(1+s^2)^2 - s^4}{s(1+s^2)} I_1$$

$$V_1 = \frac{1+\cancel{s^4}+2s^2 - \cancel{s^4}}{s+s^3} I_1$$

$$\boxed{Y_{11} = \frac{I_1}{V_1} = \frac{s+s^3}{1+2s^2}}$$

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put $V_1 = 0$

$$-\frac{1+s^2}{s} I_1 = s I_2$$

$$\boxed{I_2 = -\frac{1+s^2}{s^2} I_1}$$

$$\boxed{I_1 = -\frac{s^2}{1+s^2} I_2}$$

subs I_1 in eqn (4)

$$V_2 = s\left(\frac{-s^2}{1+s^2}\right) I_2 + \left(\frac{1+s^2}{s^2}\right) I_2$$

$$V_2 = -\frac{s^3}{1+s^2} I_2 + \frac{1+s^2}{s^2} I_2$$

$$V_2 = \frac{-s^4 + (1+s^2)^2}{s+s^3} I_2$$

$$V_2 = \frac{-\cancel{s^4} + 1 + \cancel{s^4} + 2s^2}{s+s^3} I_2$$

$$V_2 = \frac{1+2s^2}{s+s^3} I_2$$

$$\boxed{y_{22} = \frac{I_2}{V_2} = \frac{s+s^3}{1+2s^2}}$$

subs I_2 in eqn (3)

$$V_2 = \frac{1+s^2}{s} I_1 + s\left(-\frac{\cancel{1+}s^2}{\cancel{1+}s^2}\right) I_1$$

$$V_2 = \left(\frac{1+s^2}{s} - \frac{s^3}{\cancel{1+}s^2}\right) I_1$$

$$V_2 = \frac{(1+s^2)^2 - s^4}{s^3} I_1$$

$$V_2 = -\frac{\cancel{s^4} - 1 - \cancel{s^4} - 2s^2}{s^3} I_1$$

$$\boxed{\frac{I_1}{V_2} = \frac{-s^3}{1+2s^2}}$$

$$Y = \begin{bmatrix} \frac{s+s^3}{1+2s^2} & \frac{-s^3}{1+2s^2} \\ \frac{-s^3}{1+2s^2} & \frac{s+s^3}{1+2s^2} \end{bmatrix}$$

→ h parameter

$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \ldots (1)$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \ldots (2)$$

$$V_1 = \frac{1+s^2}{s} I_1 + s I_2 \quad \ldots (3)$$

$$V_2 = s I_1 + \frac{1+s^2}{s} I_2 \quad \ldots (4)$$

from (4)

$$I_2 = \frac{V_2 - s I_1}{\frac{1+s^2}{s}} = \cancel{s V_2}$$

$$I_2 = \frac{s}{1+s^2} V_2 - \frac{s^2}{1+s^2} I_1 \quad \ldots (5)$$

comparing (5) & (2) we get

$$\boxed{h_{21} = \frac{-s^2}{1+s^2}} \qquad \boxed{h_{22} = \frac{s}{1+s^2}}$$

subs I_2 in (3)

$$V_1 = \frac{1+s^2}{s} I_1 + s\left(\frac{s}{1+s^2} V_2 - \frac{s^2}{1+s^2} I_1\right)$$

$$V_1 = \frac{1+s^2}{s} I_1 + \left(\frac{s^2}{1+s^2} V_2 - \frac{s^3}{1+s^2}\right) I_1$$

$$V_1 = \frac{s^2}{1+s^2} V_2 + \left(\frac{1+s^2}{s} - \frac{s^3}{1+s^2}\right) I_1$$

$$V_1 = \frac{s^2}{1+s^2} V_2 + \left(\frac{1+\cancel{s^4}+2s^2-\cancel{s^4}}{s+s^3}\right) I_1$$

$$V_1 = \frac{s^2}{1+s^2} V_2 + \frac{1+2s^2}{s+s^3} I_1 \quad \text{(6)}$$

camparing (6) in (B)

$$\boxed{h_{11} = \frac{1+2s^2}{s+s^3}} \qquad \boxed{h_{12} = \frac{s^2}{1+s^2}}$$

$$[h] = \begin{bmatrix} \frac{1+2s^2}{s+s^3} & \frac{s^2}{1+s^2} \\ \frac{-s^2}{1+s^2} & \frac{s}{1+s^2} \end{bmatrix}$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{z_{11}}{z_{21}} & \frac{\Delta z}{z_{21}} \\ \frac{1}{z_{21}} & \frac{z_{22}}{z_{21}} \end{bmatrix}$$

$$\Delta z = \frac{1+s^2}{s}\left(\frac{1+s^2}{s}\right) - s(s)$$

$$\Delta z = \left(\frac{1+s^2}{s}\right)^2 - s^2$$

$$\Delta z = \frac{\cancel{s^4} 1 + 2s^2 - \cancel{s^4}}{s^2} \qquad \Delta z = \frac{1+2s^2}{s^2}$$

$$A = \frac{z_{11}}{z_{21}} = \frac{1+s^2}{s^2}$$

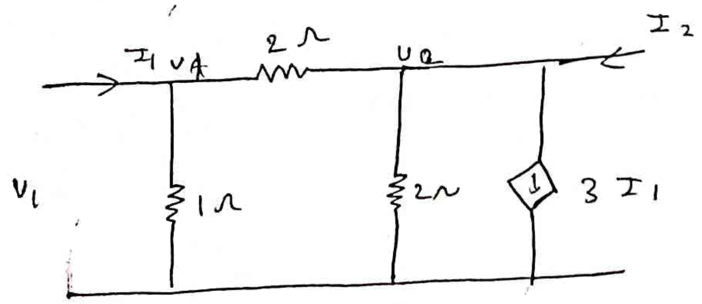
$$B = \frac{\Delta z}{z_{21}} = \frac{1+s^2}{s^3}$$

$$C = \frac{1}{z_{21}} = \frac{1}{s}$$

$$D = \frac{z_{22}}{z_{21}} = \frac{1+s^2}{s^2}$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{1+s^2}{s^2} & \frac{1+s^2}{s^3} \\ \frac{1}{s} & \frac{1+s^2}{s^2} \end{bmatrix}$$

* Find z and y parameter.



Y parameter

$$I_1 = Y_{11}V_1 + Y_{12}V_2 \quad \ldots \text{①}$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2 \quad \ldots \text{②}$$

apply KCL at node ①

$$V_A - V_2 + V_A + \frac{V_A - V_B}{2} \cancel{+\frac{V_B}{3}} = I_1$$

$$I_1 = V_1 + 0.5V_1 + 0.5V_2$$

$$I_1 = 1.5V_1 - 0.5V_2 \quad \ldots \text{③}$$

apply KCL at node ②

$$I_2 = \frac{V_2 - V_1}{2} + \frac{V_2}{2}$$

$$I_2 = 0.5V_2 - 0.5V_1 + 0.5V_2 + 3(1.5V_1 - 0.5V_2)$$

$$I_2 = -0.5V_1 + 4V_2 \quad \ldots \text{④}$$

comparing ③ & ①

$Y_{11} = 1.5$ $\quad$ $Y_{12} = -0.5$

comparing ④ & ②

$Y_{22} = -0.5$ $\quad$ $Y_{21} = 4$

$$y = \begin{bmatrix} 1.5 & -0.5 \\ 4 & -0.5 \end{bmatrix}$$

$$\Delta y = 1.5(-0.5) - (-0.5)(4)$$

$$\boxed{\Delta y = 1.25}$$

$$z_{11} = \frac{y_{22}}{\Delta y} = \frac{-0.5}{1.25} = -0.4\,\Omega$$

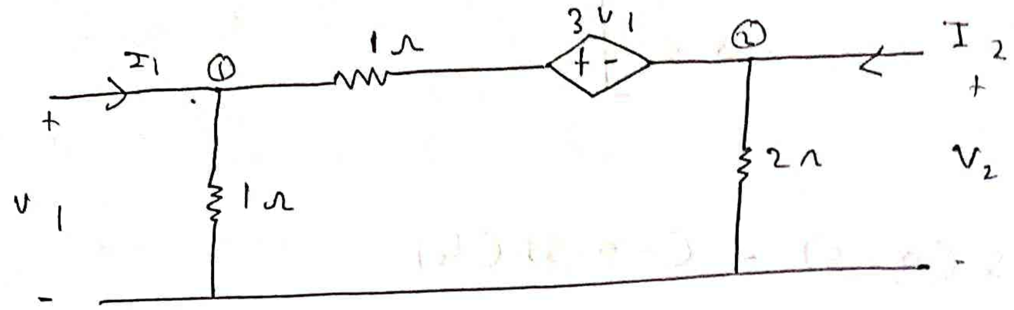
$$z_{12} = -\frac{y_{12}}{\Delta y} = -\frac{(-0.5)}{1.25} = 0.4\,\Omega$$

$$z_{22} = \frac{y_{11}}{\Delta y} = \frac{1.5}{1.25} = 1.2\,\Omega$$

$$z_{21} = \frac{-y_{21}}{\Delta y} = \frac{-4}{1.25} = -3.2\,\Omega$$

$$z = \begin{bmatrix} -0.4 & 0.4 \\ -3.2 & 1.2 \end{bmatrix}$$

* Find y and z parameters for the ckt shown.



apply KCL at ①

$$I_1 = \frac{V_1}{1} + \frac{V_1 - 3V_1 - V_2}{1}$$

$$I_1 = V_1 + V_1 - 3V_1 - V_2$$

$$\boxed{I_1 = -1V_1 - 1V_2} \quad \ldots ①$$

apply KCL at ②

$$I_2 = \frac{V_2}{2} + \frac{V_2 + 3V_1 - V_1}{1}$$

$$I_2 = (0.5 + 1)V_2 + 2V_1$$

$$\boxed{I_2 = 1.5V_2 + 2V_1} \quad \ldots ②$$

y parameters

$$I_1 = Y_{11}V_1 + Y_{12}V_2 \quad \ldots ③$$
$$I_2 = Y_{21}V_1 + Y_{22}V_2 \quad \ldots ④$$

comparing ③ & ①

$Y_{11} = -1$ $\quad\quad$ $Y_{12} = -1$

comparing ④ & ②

$Y_{21} = 2$ $\qquad$ $Y_{22} = 1.5$

$$\therefore [Y] = \begin{bmatrix} -1 & -1 \\ 2 & 1.5 \end{bmatrix}$$

$$[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$$

$$Z_{11} = \frac{Y_{22}}{\Delta y} = \frac{1.5}{-0.5} = -3\Omega$$

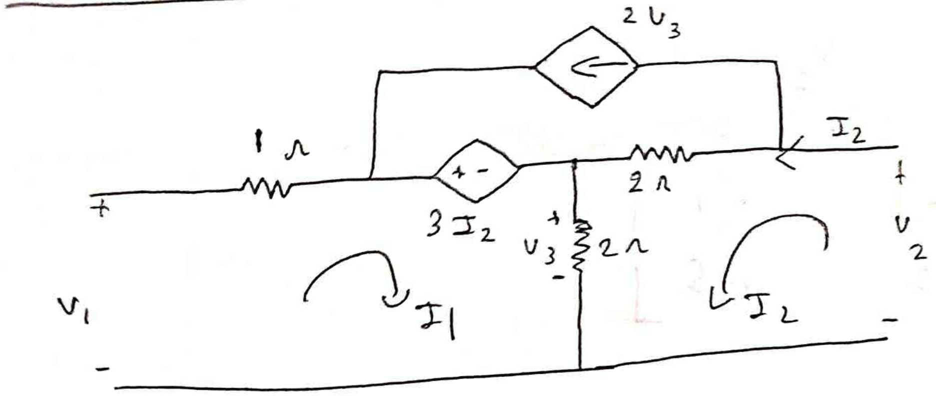
$$Z_{12} = \frac{-Y_{12}}{\Delta y} = \frac{-(-1)}{-0.5} = -2\Omega$$

$$Z_{21} = \frac{-Y_{21}}{\Delta y} = \frac{2}{0.5} = 4\Omega$$

$$Z_{22} = \frac{y_{11}}{\Delta y} = \frac{1}{0.5} = 2\Omega$$

$$[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} -3 & -2 \\ 4 & 2 \end{bmatrix}$$

* Find y & z for the circuit shown:



from ckt $I_3 = 2V_3$

$$V_3 = (I_1 + I_2)2$$

$$V_3 = 2I_1 + 2I_2$$

apply KVL to loop ①

$$V_1 - 1I_1 - 3I_2 + (2I_1 + 2I_2)$$

$$V_1 = 3I_1 + 5I_2 \quad \ldots ①$$

apply KVL to loop ②

$$V_2 - 2(I_2 - I_3) - 2(I_1 + I_2) = 0$$

$$V_2 = -2I_2 + 2I_3 - 2I_1 - 2I_2 = 0$$

$$V_2 = -2I_2 + 2(2(2I_1 + 2I_2)) - 2I_1 - 2I_2 = 0$$

$$V_2 = -2I_2 + 8I_1 + 8I_2 - 2I_1 - 2I_2 = 0$$

$$V_2 = -6I_1 - 4I_2 \quad \ldots ②$$

z params.

$$V_1 = z_{11}I_1 + z_{12}I_2 \quad \ldots ③$$

$$V_2 = z_{21}I_1 + z_{22}I_2 \quad \ldots ④$$

$z_{11} = 3\,\Omega$ $z_{12} = 5\,\Omega$

$z_{21} = -6\,\Omega$ $z_{22} = -4\,\Omega$

$$z = \begin{bmatrix} 3 & 5 \\ -6 & -4 \end{bmatrix} \Omega$$

y parameter

$$y = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} = \begin{bmatrix} \frac{z_{22}}{\Delta z} & -\frac{z_{12}}{\Delta z} \\ -\frac{z_{21}}{\Delta z} & \frac{z_{11}}{\Delta z} \end{bmatrix}$$

$$\Delta z = 3(-4) - (5(-6))$$

$$\Delta z = -12 + 30$$

$$\boxed{\Delta z = 18}$$

$$y_{11} = \frac{z_{22}}{\Delta z} = \frac{-4}{18} = -0.22\ \mho$$

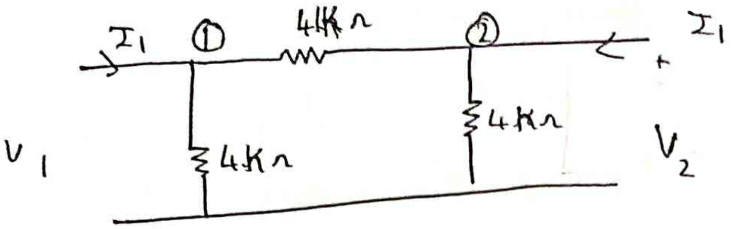
$$y_{12} = \frac{-z_{12}}{\Delta z} = \frac{-5}{18} = -0.27\ \mho$$

$$y_{21} = \frac{-z_{21}}{\Delta z} = \frac{-6}{18} = -0.33\ \mho$$

$$y_{22} = \frac{z_{11}}{\Delta z} = \frac{3}{18} = 0.166\ \mho$$

$$y = \begin{bmatrix} -0.22 & -0.27 \\ -0.33 & 0.166 \end{bmatrix}$$

* Find y parameter for the ckt shown and use them to find transmission parameter.



$$\begin{bmatrix} 0.5m & -0.25m \\ -0.25m & 0.5m \end{bmatrix}$$

$I_1 = Y_{11} V_1 + Y_{12} V_2 \quad \ldots ①$

$I_2 = Y_{21} V_1 + Y_{22} V_2 \quad \ldots ②$

apply KCL at node ①

$$I_1 = \frac{1}{4K} V_1 + \frac{V_1 - V_2}{4K}$$

$$I_1 = \left(\frac{1}{4K} + \frac{1}{4K}\right) V_1 - 0.25m V_2$$

$$I_1 = \frac{2}{4K} V_1 - 0.25m V_1$$

$$\boxed{I_1 = 0.5m V_1 - 0.25m V_2} \quad \ldots ③$$

apply KCL at node ②

$$I_2 = \frac{V_2}{4Ω} + \frac{V_2 - V_1}{4K}$$

$$I_2 = -\frac{V_1}{4K} + \frac{2}{4K} V_2 \quad \ldots$$

$$\boxed{I_2 = -0.25m V_1 + 0.5m V_2} \quad \ldots ④$$

comparing ① & ③, ② & ④

$$y = \begin{bmatrix} 0.5\,m & -0.25\,m \\ -0.25\,m & 0.5\,m \end{bmatrix}$$

$$A = -\frac{y_{22}}{y_{21}} = \frac{-0.5\,m}{-0.25\,m} = +2$$

$$B = -\frac{1}{y_{21}} = \frac{-1}{-0.25\,m} = 4K$$

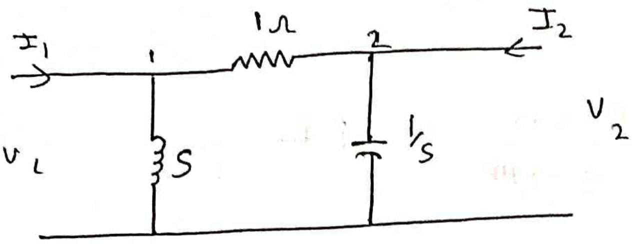
$$C = -\frac{\Delta y}{y_{21}} = \frac{-0.1875\,\cancel{m}}{-0.125\,\cancel{m}} = 0.75$$

$$D = -\frac{y_{11}}{y_{21}} = \frac{-0.5\,m}{-0.25\,m} = +2$$

$$[ABCD] = \begin{bmatrix} +2 & 4K \\ 0.75 & +2 \end{bmatrix} //$$

---

* Determine transmission parameters for the n/w shown.



apply KCL at ①

$$\frac{V_1}{s} + \frac{V_1 - V_2}{1} = I_1$$

$$I_1 = \frac{1}{s} V_1 + 1V_1 - 1V_2$$

$$I_1 = 1 + \frac{1}{s} V_1 - 1V_2$$

$$I_1 = \frac{s+1}{s} V_1 - 1V_2 \quad \dots ①$$

Apply KCL at node ②

$$I_2 = \frac{V_2}{\frac{1}{s}} + V_2 - V_1$$

$$I_2 = sV_2 + V_2 - V_1$$

$$\boxed{V_1 = (1+s)V_2 - 1I_2} \quad \dots ②$$

sub ② in ①

$$I_1 = \frac{s+1}{s}\left[(1+s)V_2 - 1I_2\right] - 1V_2 = \frac{(s+1)^2}{s}V_2 - \left(\frac{s+1}{s}\right)I_2$$

$$I_1 = \frac{s^2+1+2s}{s} - 1 \; V_2 - \left(\frac{s+1}{s}\right)I_2 = \boxed{\frac{s^2+1+s}{s}V_2 - \frac{s+1}{s}}$$

$$\boxed{I_1 = \frac{s^2+1+s}{s}V_2 - \frac{s+1}{s}I_2} \quad \dots ③$$

ABCD parameters:

$$V_1 = AV_2 - BI_2 \quad \dots ④$$

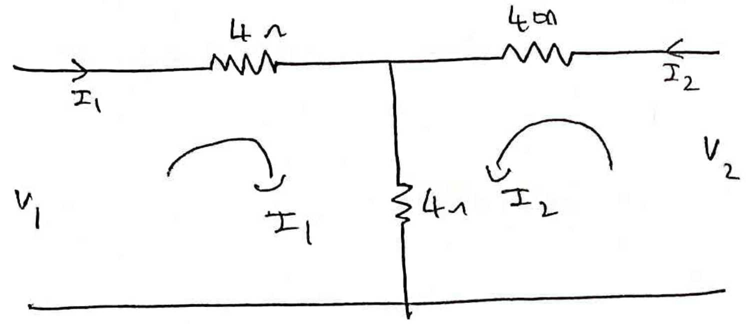
$$I_1 = CV_2 - DI_2 \quad \dots ⑤$$

comparing ④ & ②

comparing ⑤ & ③

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} s+1 & 1 \\ \frac{s^2+s+1}{s} & \frac{s+1}{s} \end{bmatrix}$$

* Determine H parameter for the ckt.



apply KVL at loop ①

$$V_1 - 4I_1 - 4I_1 - 4I_2 = 0$$

$$\boxed{V_1 = 8I_1 + 4I_2} \quad \ldots ①$$

apply KVL to loop ②

$$V_2 - 4I_2 - 4I_2 - 4I_1 = 0$$

$$\boxed{V_2 = 8I_2 + 4I_1} \quad \ldots ②$$

$$I_2 = \frac{V_2 - 4I_1}{8}$$

$$\boxed{I_2 = 0.125V_2 - 0.5I_1} \quad \ldots ③$$

subs I_2 in ①

$$V_1 = 8I_1 + 4(0.125V_2 - 0.5I_1)$$

$$V_1 = 8I_1 + 0.5V_2 - 2I_1$$

$$\boxed{V_1 = 6I_1 + 0.5V_2} \quad \ldots ④$$

h parameters

$$V_1 = h_{11}I_1 + h_{12}V_2 \quad \ldots ⑤$$

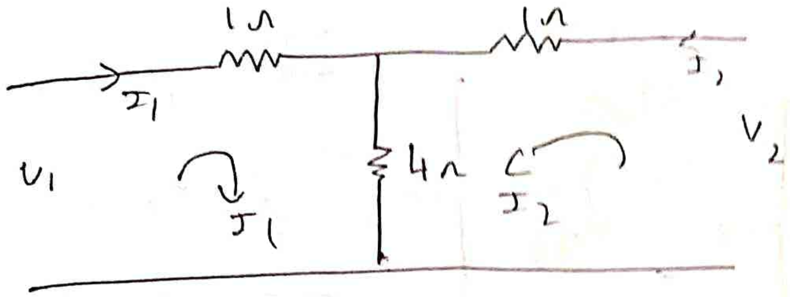
$$I_2 = h_{21}I_1 + h_{22}V_2 \quad \ldots ⑥$$

comparing ④ & ⑤

comparing ③ & ⑥

$$[h] = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} 6 & 0.5 \\ -0.5 & 0.125 \end{bmatrix}$$

Determine A B C D parameters from the ckt.



ABCD parameters

$V_1 = AV_2 - BI_2 \quad \ldots (1)$

$I_1 = CV_2 - DI_2 \quad \ldots (2)$

apply KVL to loop (1),

$V_1 - I_1 - 4(I_1 + I_2) = 0$

$V_1 - 5I_1 - 4I_2 = 0$

$\boxed{V_1 = 5I_1 + 4I_2} \quad \ldots (3)$

$V_2 - I_2 - 4(I_1 + I_2)$

$V_2 - 5I_2 = 4I_1$

$\boxed{I_1 = 0.25V_2 - 1.25I_2} \quad \ldots (4)$

subs (4) in (3)

$V_1 = 5(0.25V_2 - 1.25I_2) + 4I_2$

$V_1 = 1.25V_2 - 6.25I_2 + 4I_2$

$\boxed{V_1 = 1.25V_2 - 2.25I_2} \quad \ldots (5)$

comparing (5) & (1)

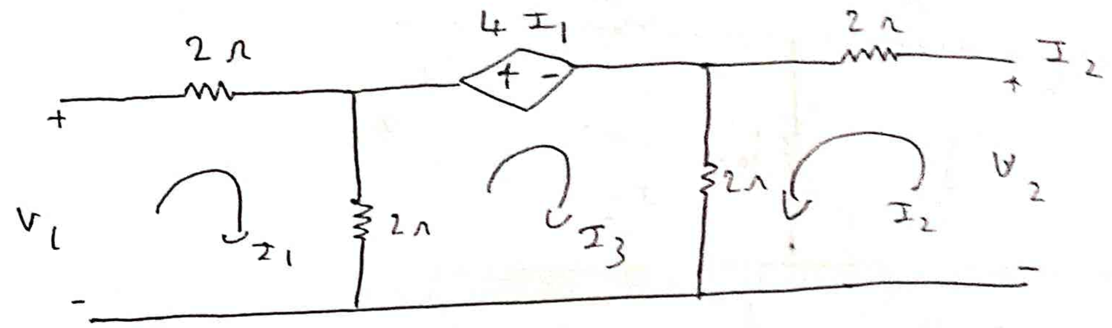
$A = 1.25 \qquad B = 2.25$

comparing (4) & (2)

$C = 0.25 \qquad D = -1.25$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1.25 & 2.25 \\ 0.25 & -1.25 \end{bmatrix}$$

→ find Z & h parameters for the circuit shown



apply KVL to loop ①

$$V_1 - 2I_1 - 2I_1 + 2I_3 = 0$$

$$\boxed{V_1 = 4I_1 - 2I_3}$$

apply KVL to loop ②

$$-2I_2 - 2(I_2 + I_3) + V_2 = 0$$

$$-2I_2 - 2I_2 - 2I_3 + V_2 = 0$$

$$\boxed{V_2 = +4I_2 + 2I_3}$$

apply KVL to loop ③

$$-4I_1 - 2(I_2 + I_3) - 2(I_3 - I_1) = 0$$

$$-4I_1 - 2I_2 - 2I_3 - 2I_3 + 2I_1 = 0$$

$$\frac{-2I_1 - 2I_2 - 4I_3 = 0}{2}$$

$$-I_1 - I_2 - 2I_3 = 0$$

$$\boxed{2I_3 = -I_1 - I_2}$$

subs $2I_3$ in V_1

$$V_1 = 4I_1 - (-I_1 - I_2)$$

$$\boxed{V_1 = 5I_1 + 1I_2}$$

subs $2I_3$ in V_2

$$V_2 = +4I_2 + (-I_1 - I_2)$$

$$\boxed{V_2 = -1I_1 + 3I_2}$$

$$Z = \begin{bmatrix} 5 & 1 \\ -1 & 3 \end{bmatrix}$$

<u>h parameters:</u>

$$[h] = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{\Delta Z}{Z_{22}} & \frac{Z_{12}}{Z_{22}} \\ \frac{-Z_{21}}{Z_{22}} & \frac{1}{Z_{22}} \end{bmatrix}$$

$$h_{11} = \frac{\Delta Z}{\Delta Z_{22}} = \frac{16}{3} = 5.33$$

$$h_{12} = \frac{Z_{12}}{Z_{22}} = \frac{1}{3} = 0.33$$

$$h_{21} = \frac{-Z_{21}}{Z_{22}} = \frac{+1}{3} = 0.33$$

$$h_{22} = \frac{1}{Z_{22}} = \frac{1}{3} = 0.33$$

$$\therefore [h] = \begin{bmatrix} 5.33 & 0.33 \\ 0.33 & 0.33 \end{bmatrix}$$

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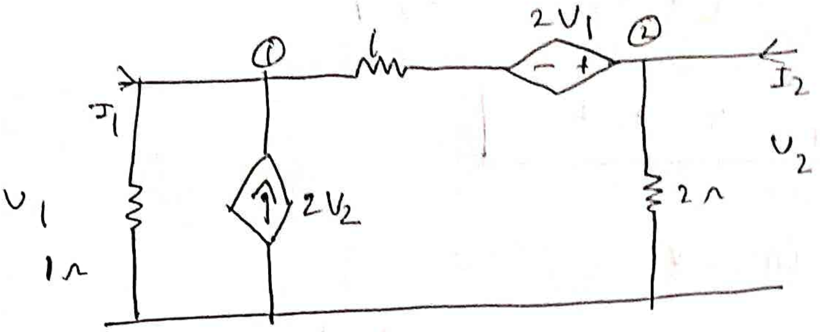
→ Find y and z parameters for the n/w shown.



apply KCL at ①

$$V_1 + \underbrace{\frac{V_1 + 2V_1 - V_2}{1}} = I_1 + 2V_2$$

$$\boxed{4V_1 - 3V_2 = I_1}$$

apply kcl at ②

$$\frac{V_2}{2} + V_2 - 2V_1 - V_1 = I_2$$

$$\boxed{1.5V_2 - 3V_1 = I_2}$$

~~put $V_2 = 0$~~

$y_{11} = \frac{I_1}{V_1} = 4$ $\qquad y_{12} = \frac{I_1}{V_2} = -3$

$y_{21} = -3$ $\qquad y_{22} = 1.5$

$$Y = \begin{bmatrix} 4 & -3 \\ -3 & 1.5 \end{bmatrix} \mho$$

$$z = \begin{bmatrix} \frac{1.5}{-3} & -\left(\frac{-3}{-3}\right) \\ -\left(\frac{-3}{-3}\right) & \frac{+4}{-3} \end{bmatrix}$$

$$Z = \begin{bmatrix} -0.5 & -1 \\ -1 & 1.33 \end{bmatrix} \Omega$$

→ Prove that $AD - BC = 1$ for a reciprocal n/w

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{Z_{11}}{Z_{21}} & \frac{\Delta Z}{Z_{21}} \\ \frac{1}{Z_{21}} & \frac{Z_{22}}{Z_{21}} \end{bmatrix}$$

$A = \frac{Z_{11}}{Z_{21}}$ $\quad B = \frac{\Delta Z}{Z_{21}}$ $\quad C = \frac{1}{Z_{21}}$ $\quad D = \frac{Z_{22}}{Z_{21}}$

$$\Delta Z = Z_{11} Z_{22} - Z_{12} Z_{21}$$

$$= \begin{bmatrix} \frac{Z_{11}}{Z_{21}} & \frac{Z_{11} Z_{22} - Z_{12} Z_{21}}{Z_{21}} \\ \frac{1}{Z_{21}} & \frac{Z_{22}}{Z_{21}} \end{bmatrix}$$

$$= \frac{Z_{11} Z_{22}}{Z_{21}^2} - \frac{Z_{11} Z_{22} - Z_{21} Z_{12}}{Z_{21}^2}$$

$$= \frac{\cancel{Z_{11} Z_{22}} - \cancel{Z_{11} Z_{22}} + Z_{21} Z_{12}}{Z_{21}^2}$$

for a reciprocal n/w $Z_{21} = Z_{12}$

$\therefore Z_{12} = Z_{21}$

$$= \frac{\cancel{Z_{21}^2}}{\cancel{Z_{21}^2}}$$

$$\boxed{\therefore AD - BC = 1}$$

(18) For a symmetrical n/w show that $\Delta h = 1$

Soln:

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{\Delta Z}{Z_{22}} & \frac{Z_{12}}{Z_{22}} \\ \frac{-Z_{21}}{Z_{22}} & \frac{1}{Z_{22}} \end{bmatrix}$$

$$\Delta h = \frac{\Delta Z}{Z_{22}^2} - \frac{Z_{12}(-Z_{21})}{Z_{22}^2}$$

$$\Delta h = \frac{\Delta Z}{Z_{22}^2} + \frac{Z_{12} Z_{21}}{Z_{22}^2}$$

$$\Delta h = \frac{\Delta Z + Z_{12} Z_{21}}{Z_{22}^2}$$

$$\Delta Z = Z_{11} Z_{22} - Z_{12} Z_{21}$$

$$\Delta h = \frac{\cancel{Z_{12} Z_{21}} + Z_{11} Z_{22} - \cancel{Z_{12} Z_{21}}}{Z_{22}^2}$$

for a symetrical n/w $Z_{11} = Z_{22}$

$$\therefore Z_{22} = Z_{11}$$

$$\Delta h = \frac{Z_{22} \cdot Z_{22}}{Z_{22}^2}$$

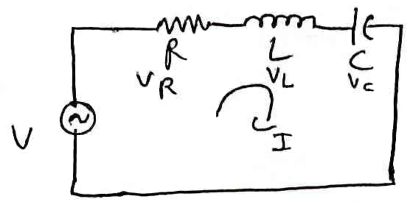
$$\Delta h = \frac{\cancel{Z_{22}^2}}{\cancel{Z_{22}^2}}$$

$$\boxed{\Delta h = 1}$$

//hp

Resonance:

AC circuits which are made up of resistor, inductor, capacitor are said to be resonance circuits, when current drawn from the supply is in phase with the input voltage



→ Resonance:

It is a phenominon or study of AC ckts consisting of R, L & C of a variable frequency.

→ resonance are of two types

series & parallel resonance

current I is given by

$$I = \frac{V}{Z}$$

where Z is the impedence of ckt.

$$\boxed{Z = R + j(x_L - x_C)}$$

$$X_L = 2\pi f L$$

$$X_C = \frac{1}{2\pi f C}$$

Resonance is a ~~point~~ condition where $X_L = X_C$ for variable frequency.

at resonance $Z = R + \underbrace{j(x_L - x_C)}_{\to 0}$

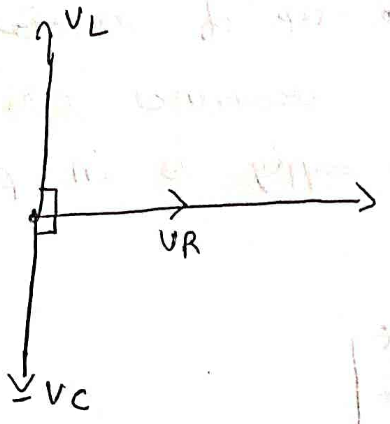
$$\boxed{Z = R}$$

Purely resistive ckt

$$\boxed{I_M = \frac{V}{R}}$$

$$\boxed{I_M = I_{max}}$$

Phasor diagram



Exp for resonant frequency.

at Resonance

$$X_L = X_C$$

$$2\pi f L = \frac{1}{2\pi f C} \implies \sqrt{f_r^2} = \frac{1}{\sqrt{4\pi^2 LC}} \quad f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$\boxed{f_r = \frac{1}{2\pi\sqrt{LC}}}$$

- resonance can be acheived by varing f_r by keeping L & C constant
- by varing L & C and keeping resonat frequency constan

→ **Features of Resonance circuits:**

- Inductive reactance = X_C (capacitive reactance)

$$X_L = X_C$$

- output current is in phase with the input volt
- capacitive voltage = - of inductive voltage
- circuit works as purely resistive circuit
- power factor is unity

$$P = VI\cos\phi \quad \text{or} \quad \cos\phi = \frac{P}{VI} = 1$$

• Maximum current & minimum impedance.

1) A coil of 5mh inductance & 10 Ω resistance is connected in series with 5 μF capacitor determine the frequency at which resonance

given :- $L = 5m$

$C = 5\mu$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$f_r = \frac{1}{2\pi\sqrt{5 \times 10^{-3} \times 5 \times 10^{-6}}}$$

$$\boxed{f_r = 1\ KHz}$$

---

2) In a series RLC ckt driven with a sinusoidal AC voltage determine the value of capacitor required to acheive resonance at 5 KHz. if $R = 2\Omega$ & $L = 1mH$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

square on both side

$$f_r^2 = \frac{1}{4\pi^2 LC}$$

$$C = \frac{1}{4\pi^2 L f_r^2}$$

$$C = \frac{1}{4\pi^2 \times 1 \times 10^{-3} \times \ldots}$$

$$\boxed{C = 1.01\ \mu F}$$

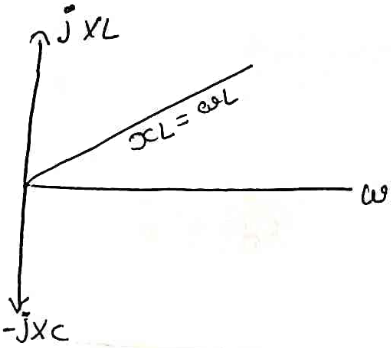
→ Reactance curve:

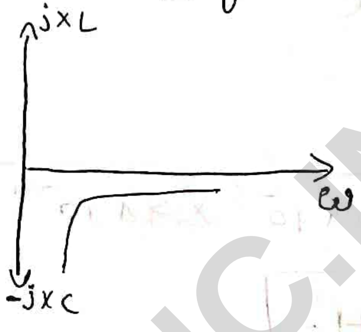
The graph of individual reactance vs the frequency

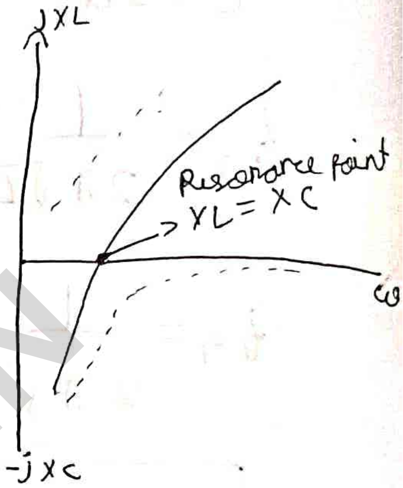
$X_L = \omega L$ $\quad\quad$ $X_C = \frac{1}{\omega C}$ $\quad\quad$ $X_L = X_C$

$X_L \propto \omega$ $\quad\quad$ $X_C \propto \frac{1}{\omega}$

$X_L \propto 2\pi f r$ $\quad\quad$ $X_C = \frac{1}{2\pi f}$







· Variation of Impedance, Admittance and current with frequency.

Let $z = R + jx_L - jX_C$ $\quad\quad$ X_L

X_C

$Z = R + j(x_L - x_C)$

$|Z| = \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}$

at resonance $X_L = x_C$

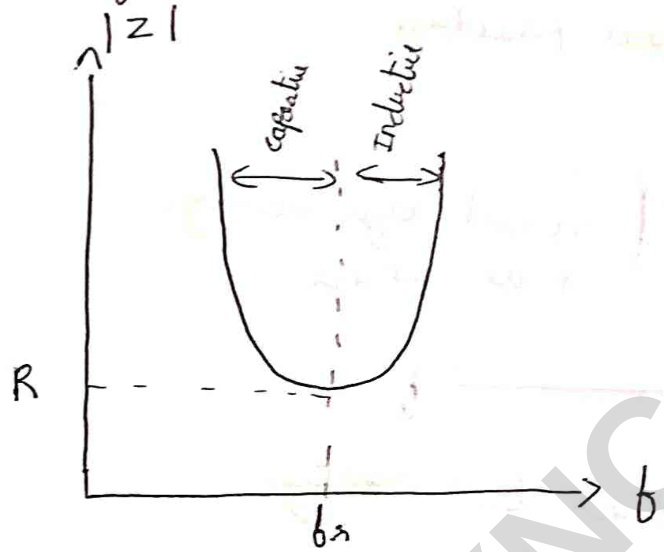
$\therefore |Z| = \sqrt{R^2}$

$\boxed{|Z| = R}$

. At Resonance, magnitude of impendance is equal to R

$$\boxed{|Z| = R}$$

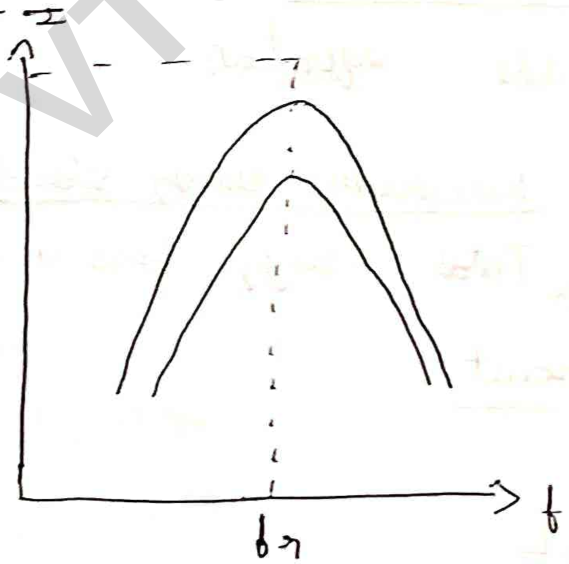
. At frequencies other than f_r, the reactances are not equal to zero and magnitude of impedance is also greater than R



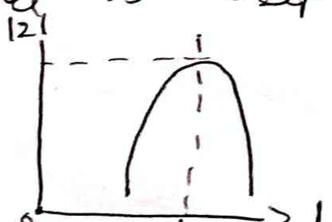
at series resonance impedance is minimum

as impedance is minimum the current flowing in the circuit

$V = I\cancel{R}$
$I = \frac{V}{Z}$



since admitance is reciprocal of Z graph is shown below.



- at frequencies below resonant frequency. $(f < f_r)$.

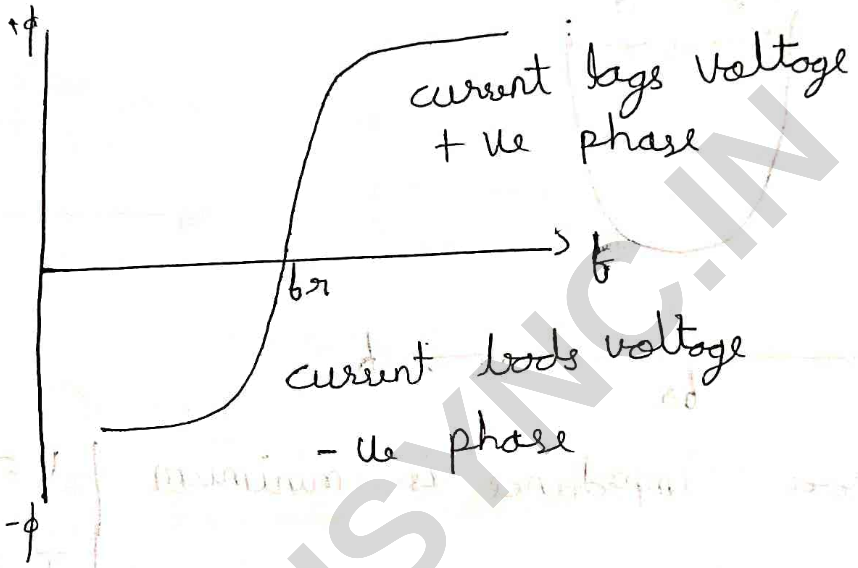
$(X_C > X_L)$ current leads the voltage.

$\therefore$ phase is ~~positive~~ Negative.

- at frequencies above resonant frequency $(f_r < f)$

$(X_L > X_C)$ current lags the voltage

$\therefore$ phase is ~~negat~~ positive

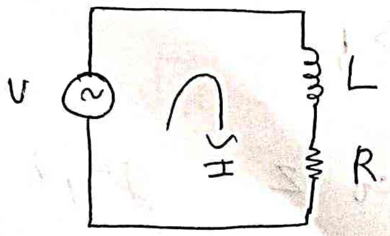


→ Q factor (Quality factor)

It is given by the equation

$$Q = 2\pi \left(\frac{\text{maximum energy stored}}{\text{Total energy loss in a period of time}} \right)$$

→ for Inductive circuit :



The maximum energy stored in the ckt is given by

$$\text{max energy} = I^2 \times L$$

$$\text{maximum energy} = \left(\frac{I_{rms}}{\sqrt{2}}\right)^2 \times L$$

$$\text{maximum energy} = \frac{I_m^2 L}{2} \quad \ldots \quad (1)$$

Total energy loss in the period is given by $P \times T$

$$\text{Total energy} = P \times T$$
$$= I^2 R \times T$$
$$= \frac{I_m^2}{2} RT \quad \ldots \quad (2)$$

(1) & (2) subs in

$$Q = 2\pi \frac{\frac{\cancel{I_m^2} L}{\cancel{2}}}{\frac{\cancel{I_m^2} RT}{\cancel{2}}}$$

$$Q = 2\pi\left(\frac{L}{RT}\right)$$

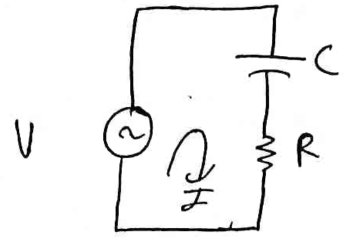
$$Q = \frac{2\pi f L}{R}$$

$$\boxed{Q = \frac{\omega L}{R}}$$

at resonance $\omega = \omega_0$

$$\boxed{Q_0 = \frac{\omega_0 L}{R}}$$

→ for capacitive circuit



consider of Q of inductive ckt

at resonance we know that

$X_L = X_C$

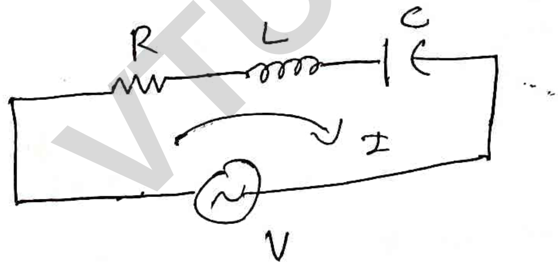
$\omega_0 L = \frac{1}{\omega_0 C}$

WKT

$\therefore Q_0 = \frac{\omega_0 L}{R}$

$$\boxed{\therefore Q_0 = \frac{1}{\omega_0 C R}}$$

→ Q factor for series RLC ckt



consider $Q = \frac{\omega_0 L}{R}$ ... (1)

WKT

$\omega_0 = 2\pi f_r$

$f_r = \frac{1}{2\pi\sqrt{LC}}$

$2\pi f_r = \frac{1}{\sqrt{LC}}$ ... (2)

sub (2) in (1)

$$q = \frac{L}{\sqrt{LC}\, R}$$

$$q = \frac{1}{\cancel{\sqrt{L}}} \frac{\cancel{\sqrt{L}}}{\sqrt{C}} \quad \sqrt{L} \times \frac{1}{R}$$

$$q = \left(\sqrt{\frac{L}{C}}\right)\frac{1}{R}$$

$$\boxed{q = \frac{1}{R}\left(\sqrt{\frac{L}{C}}\right)}$$

→ series resonance ckt as voltage follower.

voltage across L & C under resonance

· For inductive circuit.

WKT

$$I_0 = \frac{V}{Z}$$

at resonance $Z = R$

$$I_0 = \frac{V}{R}$$

$\boxed{V_L = I_0\, jX_L}$ voltage across inductor

$$V_L = \frac{V}{R} jX_L$$

$$V_L = \frac{V}{R} j\omega L$$

$$V_L = j\left(\frac{\omega L}{R}\right) V$$

WKT $\frac{\omega L}{R} = q_0$.

$$\boxed{V_L = j\, q_0\, V}$$

q_0 = amplification factor

$\boxed{V_L = q_0 V \angle 90}$ ...... ① leads by 90^0

For capacitive ckt:

$$I_0 = \frac{V}{Z}$$

at resonance.

$$I_0 = \frac{V}{R}$$

$$V_C = I_0(-jX_C)$$

$$V_C = \frac{V}{R} \, -j\frac{1}{\omega C}$$

$$V_C = -j\left(\frac{1}{\omega CR}\right)V$$

we know that $\frac{1}{\omega CR} = Q_0$

$$V_C = -j(Q_0)V$$

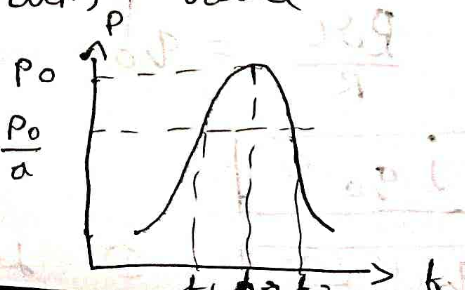
$$\boxed{V_C = Q_0 V \angle -90^\circ} \quad \ldots (2)$$ voltage lags by 90°

at resonance, from the above exp it is understood that V_L & V_C have same magnitude but they are out of phase hence both gets cancld at Resonance. The magnitude of V_L and V_C are Q_0 times the input voltage V, hence at resonance series RLC circuit works like voltage amplifier where Q_0 is the amplification factor.

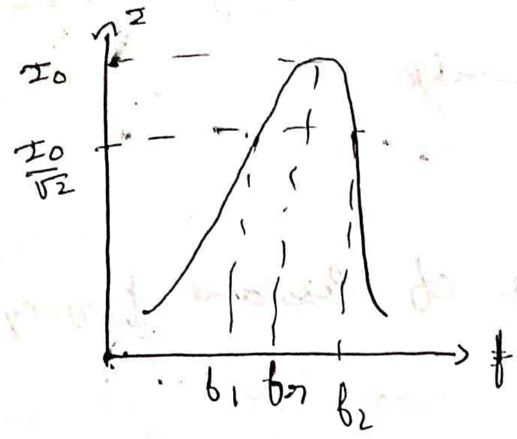
Bandwidth and selectivity of series resonance ckt:

Defined as Band of frequency the powers in the ckt is half of its maximum value

$$BW = f_2 - f_1$$



In terms of current



general power equation:

$$P = I^2 R$$

The exp for maximum power is

$$\boxed{P_{max} = I_{max}^2 R}$$

WKT,

$$I_{max} = \frac{I_m}{\sqrt{2}}$$

$$P_{\cancel{max}} = \frac{I_m^2}{2} R$$

$$\boxed{P_{\cancel{max}} = \frac{P_{max}}{2}}$$

The frequencies at which power in the ckt is half of its maximum value are called <u>half power frequencies</u>.

→ Selectivity:

Defined as ability of ckt to distinguish b/w desired and undesired frequency

or

also defined as the Ratio of Resonance frequency to the BW of resonance ckt

$$S = \frac{f_r}{BW}$$

$BW = \frac{R}{L}$

$BW = \frac{R}{2\pi L}$

$$\boxed{S = \frac{f_r}{f_2 - f_1}}$$

$BW = f_2 - f_1$

$$\boxed{S \propto \frac{1}{BW}}$$

Selectivity is given by.

$$S = \frac{f_r}{BW}$$

$$S = \frac{f_r}{R/2\pi L}$$

$$S = \frac{2\pi L f_r}{R}$$

$$S = \frac{\omega_0 L}{R}$$

$$\boxed{S = Q_0}$$

$$\boxed{S \propto Q_0}$$

$$\boxed{Q_0 \propto \frac{1}{BW}}$$

as Q_0 increases Bandwidth decreases and vice versa.

***

. Relatation ships b/w resonance f & half power frequency.

or

Derive the exp. for half power / cutoff / 3dB frequency

or

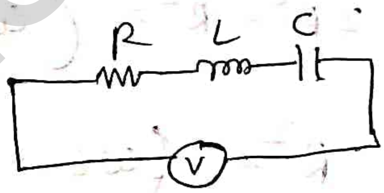
Exp for half power frequency in terms of BW etc.

in a series RLC ckt:

Let f_0 be the resonant frequency of series RLC ckt and f_1 & f_2 be the half power frequency at which current is $\frac{1}{\sqrt{2}}$ times of I_0

current in series RLC ckt is given by.

$$I = \frac{V}{|Z|}$$



we know that

$$Z = R + j(X_L - X_C)$$

$$|Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$\therefore I = \frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \quad \ldots \text{(1)}$$

current at half power frequency is given by

$$I = \frac{I_0}{\sqrt{2}}$$

$$I = \frac{1}{\sqrt{2}} \frac{V}{Z} \quad \ldots \text{(2)}$$

subs ② in ①

$$\frac{1}{\sqrt{2}}\frac{V}{|Z|} = \frac{V}{\sqrt{R^2+\left(\omega L-\frac{1}{\omega C}\right)^2}}$$

at resonance

$$\frac{1}{\sqrt{2}}\cdot\frac{1}{R} = \frac{1}{\sqrt{R^2+\left(\omega L-\frac{1}{\omega C}\right)^2}}$$

$$R\sqrt{2} = \sqrt{R^2+\left(\omega L-\frac{1}{\omega C}\right)^2}$$

square on both sides,

$$2R^2 = R^2+\left(\omega L-\frac{1}{\omega C}\right)^2$$

$$2R^2-R^2 = \left(\omega L-\frac{1}{\omega C}\right)^2$$

$$R^2 = \left(\omega L-\frac{1}{\omega C}\right)^2$$

$$\boxed{\omega L-\frac{1}{\omega C} = \pm R} \quad \cdots ③$$

Let $\boxed{\left(\omega_1 L-\frac{1}{\omega_1 C}\right) = -R} \quad \cdots ④$

$$\boxed{\left(\omega_2 L-\frac{1}{\omega_2 C}\right) = +R} \quad \cdots ⑤$$

case ① adding 4 & ⑤

$$\left(\omega_1 L-\frac{1}{\omega_1 C}\right)+\left(\omega_2 L-\frac{1}{\omega_2 C}\right) = 0$$

$$\omega_1 L + \omega_2 L - \left(\frac{1}{\omega_1 C} + \frac{1}{\omega_2 C}\right) = 0$$

$$L(\cancel{\omega_1 + \omega_2}) = \frac{1}{C}\left(\frac{\cancel{\omega_2 + \omega_1}}{\omega_1 \omega_2}\right)$$

$$\therefore \boxed{\omega_1 \omega_2 = \frac{1}{LC}} \quad \dots (6)$$

WKT $\omega_0 = \frac{1}{\sqrt{LC}}$

Square on both sides

$$\omega_0^2 = \frac{1}{LC} \quad \dots (7)$$

from (6) & (7)

$$\boxed{\omega_1 \omega_2 = \omega_0^2}$$

$$2\pi f_1 + 2\pi f_2 = (2\pi f_0)^2$$

$$\boxed{\cancel{4\pi^2}(f_1 f_2) = \cancel{4\pi^2} f_0^2}$$

$$f_0^2 = f_1 f_2$$

$$\boxed{f_0 = \sqrt{f_1 f_2}} \quad \dots (A)$$

show that the geometric mean of two half power frequency is equal to f_0 (Resonating frequency).

Case 2: subtracting eqn (4) - (5)

$$\left(\omega_1 L - \frac{1}{\omega_1 C}\right) - \left(\omega_2 L - \frac{1}{\omega_2 C}\right) = -R-(+R)$$

~~x by (-1)~~

$$(\omega_1 L - \omega_2 L) - \left(\frac{1}{\omega_1 C} + \frac{1}{\omega_2 C}\right) = -2R$$

$$L(\omega_1 - \omega_2) - \left(\frac{1}{\omega_1} - \frac{1}{\omega_2}\right)\frac{1}{C} = -2R$$

$$\frac{1}{C}\left(\frac{1}{\omega_1} - \frac{1}{\omega_2}\right) + (\omega_2 - \omega_1)L = 2R$$

$$(\omega_2 - \omega_1)L + \frac{1}{C}\left(\frac{\omega_2 - \omega_1}{\omega_1 \omega_2}\right) = 2R$$

$\div L$

$$(\omega_2 - \omega_1) + \frac{1}{LC}\left(\frac{\omega_2 - \omega_1}{\omega_1 \omega_2}\right) = \frac{2R}{L}$$

$$\frac{1}{LC} = \omega_1 \omega_2$$

$$(\omega_2 - \omega_1) + \cancel{\omega_1 \omega_2}\frac{(\omega_2 - \omega_1)}{\cancel{\omega_1 \omega_2}} = \frac{2R}{L}$$

$$\cancel{2}(\omega_2 - \omega_1) = \cancel{2}\frac{R}{L}$$

$$2\pi f_2 - 2\pi f_1 = \frac{R}{L}$$

$$2\pi(f_2 - f_1) = \frac{R}{L}$$

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$$f_2 - f_1 = \frac{R}{2\pi L}$$

$$\boxed{BW = \frac{R}{2\pi L}}$$

selectivity is given by.

$$S = \frac{f_r}{BW}$$

$$S = \frac{f_r}{\frac{R}{2\pi L}}$$

$$S = \frac{\omega_0 L}{R} = Q_0$$

$$\boxed{S = Q_0}$$

- Properties of series resonant ckt

- I_0 in phase with V
- unity power factor
- at resonance impedance decreases current increases
- I ↑ → power ↑
- series RLC ckt works as voltage amplifier
- Resonance can be acheived.
  - By keeping Fr constant varying L & C
  - By keeping L & C constant & varying Fr

- $Q_0 \propto S$

- $BW \propto \frac{1}{S}$

- $Q_0 \propto \frac{1}{BW}$

→ Expression for ω_1 & ω_2

wkt

$$|Z| = \sqrt{2}R$$

$$\sqrt{R^2 + \left(\omega_L - \frac{1}{\omega C}\right)^2} = \sqrt{2}R$$

squaring on both sides

$$\sqrt{R^2 + \left(\omega L - \frac{1}{\omega L}\right)^2} = 2R^2$$

$$R^2 = \left(\omega L - \frac{1}{\omega C}\right)^2$$

$$\boxed{\omega_L - \frac{1}{\omega C} = \pm R} \quad \cdots \text{(1)}$$

$$\omega_2 L - \frac{1}{\omega_2 C} = R$$

$$\frac{\omega_2^2 LC - 1}{\omega_2 C} = R$$

$$\omega_2^2 LC - 1 = R\omega_2 C$$

$$\omega_2^2 LC - \omega_2 RC - 1 = 0 \quad \ldots \text{(2)}$$

eq (2) is the form of quadratic eqn.

hence

$$ax^2 + bx + c = 0 \quad \ldots \text{QE}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$a = LC$, $b = -RC$, $c = -1$

$$\omega_2 = \frac{-(-RC) \pm \sqrt{(-RC)^2 - (4)(LC)(-1)}}{2LC}$$

$$\omega_2 = \frac{RC \pm \sqrt{R^2C^2 + 4LC}}{2LC}$$

$$\omega_2 = \frac{R\cancel{C}}{2L\cancel{C}} \pm \sqrt{\frac{R^2C^2 + 4LC}{4L^2C^2}}$$

$$\omega_2 = \frac{R}{2L} \pm \sqrt{\frac{R^2\cancel{C^2}}{4L^2\cancel{C^2}} + \frac{\cancel{4}\cancel{L}\cancel{C}}{\cancel{4}L^{\cancel{2}}C^{\cancel{2}}}}$$

$$\boxed{\omega_2 = \frac{R}{2L} + \sqrt{\frac{R^2}{4L^2} + \frac{1}{LC}} \quad \ldots \text{(7)}}$$

Put $\omega = 1$ in (1)

$$\omega_1 L - \frac{1}{\omega_1 C} = -R$$

$$\frac{\omega_1^2 LC - 1}{\omega_1 C} = -R$$

$$\omega_1^2 LC - 1 = -R\omega_1 C$$

$$\omega_1^2 LC + \omega_1 RC - 1 = 0 \quad \ldots (2)$$

2 is of the form QE

$$a x^2 + bx + c$$

$$a = LC \quad b = RC \quad c = -1$$

$$\omega_1 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\omega_1 = \frac{-RC \pm \sqrt{R^2C^2 + 4LC}}{2LC}$$

$$\omega_1 = -\frac{R\cancel{C}}{2L\cancel{C}} \pm \sqrt{\frac{R^2C^2 + 4LC}{4L^2C^2}}$$

$$\omega_1 = -\frac{R}{2L} \pm \sqrt{\frac{R^2\cancel{C^2}}{4L^2\cancel{C^2}} + \frac{4LC}{4L^2C^2}}$$

$$\boxed{\omega_1 = -\frac{R}{2L} \pm \sqrt{\frac{R^2}{4L^2} + \frac{1}{LC}}}$$

$$\boxed{\omega_1 = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}} \quad (4)$$

Expressing eqn (3) & (4) in terms of BW

$$BW = f_2 - f_1$$

$$BW = \omega_2 - \omega_1$$

$$BW = \left[\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}\right] - \left[-\frac{R}{2L} \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}\right]$$

$$\frac{R}{2L} + \cancel{\sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}} + \frac{R}{2L} - \cancel{\sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}}$$

$$BW = \frac{\cancel{2}R}{\cancel{2}L}$$

$$\boxed{BW = \frac{R}{L}}$$

To show that $f_0 = \sqrt{f_1 f_2}$ or prove that f_0 = ~~Ge~~ Geometric mean of two half power frequency / cutoff / 3 db frequency.

$$\omega_1 \omega_2 = \omega_0^2$$

$$\left[-\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}\right] \cdot \left[\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}\right] = \omega_0^2$$

$$= -\frac{R^2}{4L^2} - \cancel{\frac{R}{2L}\sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}} + \cancel{\frac{R}{2L}\sqrt{\left(\frac{R^2}{2L}\right)^2 + \frac{1}{LC}}} + \frac{R^2}{4L} + \frac{1}{LC}$$

$$= -\cancel{\frac{R^2}{4L^2}} + \cancel{\frac{R^2}{4L}} + \frac{1}{LC}$$

$$= \frac{1}{LC} \quad \ldots (5)$$

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$\omega_0 = \frac{1}{\sqrt{LC}}$

$\omega_0^2 = \frac{1}{LC}$ .. (6)

equating 5 & (6)

$\omega_1 \omega_2 = \omega_0^2$

$\omega_0 = \sqrt{\omega_1 \omega_2}$

$2\pi f_0 = \sqrt{2\pi f_1 \, 2\pi f_2}$

$2\pi f_0 = \sqrt{4\pi^2 f_1 f_2}$

$\cancel{2\pi} f_0 = \cancel{2\pi}\sqrt{f_1 f_2}$

$$\boxed{f_0 = \sqrt{f_1 f_2}}$$

- Prove that f_0 is the arithmetic mean of ω_1 & ω_2

$$f_0 = \frac{f_1 + f_2}{2}$$

WKT

$\omega_1 = \frac{-R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} = \frac{-R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \omega_0}$

$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} = \frac{R}{2L} + \omega_0 + \sqrt{\left(\frac{R}{2L}\right)^2}$

If $\frac{R}{2L}$ is very much lesser than $\frac{1}{LC}$

$$\omega_1 = -\frac{R}{2L} + \sqrt{\frac{1}{LC}} = -\frac{R}{2L} + \omega_0$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\frac{1}{LC}} = \frac{R}{2L} + \omega_0$$

$$\omega_1 + \omega_2$$

$$-\cancel{\frac{R}{2L}} + \omega_0 + \cancel{\frac{R}{2L}}\ \omega_0$$

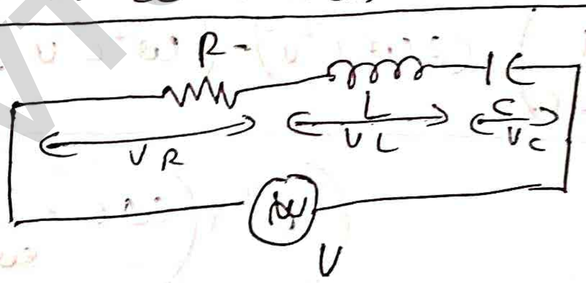
$$\omega_1 + \omega_2 = 2\omega_0$$

$$\omega_0 = \frac{\omega_1 + \omega_2}{2}$$

$$\cancel{2\pi} f_0 = \cancel{2\pi}\ \frac{f_1 + f_2}{2}$$

$$\boxed{f_0 = \frac{f_1 + f_2}{2}}$$

express of ω (ω_L) at which V_L is maximum.



$$V_L = IR$$
$$= I X_L$$

$$V_L = I\omega L \quad \cdots (1)$$

W.K.T

$$I = \frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \quad \cdots (2)$$

Sub (2) in (1)

$$V_L = \omega L \frac{V}{\sqrt{R^2 + \left(\omega_L + \frac{1}{\omega c}\right)^2}} \qquad V_L^2 = \frac{\omega^2 L^2 V^2}{R^2 + (\omega L - \frac{1}{\omega c})}$$

$\frac{dV_L}{d\omega} = 0$ for maximum voltage across inductor

$$\frac{dV_L}{d\omega} = \frac{R^2 + \left(\omega L - \frac{1}{\omega L}\right)^2 \frac{d}{d\omega} \omega^2 L^2 V^2 - \omega^2 L^2 V^2 \frac{d}{d\omega} (\omega L}{\left(R^2 + \left(\omega L - \frac{1}{\omega}\right)^2\right)^2}$$

$$-\left(\frac{1}{\omega^2}\right)$$

$$\frac{dV_L}{d\omega} = \frac{R^2 + \left(\omega L - \frac{1}{\omega c}\right)^2 2\omega L^2 V^2 - (\omega^2 L^2 V^2)\, 2\left(\omega L - \frac{1}{\omega c}\right)\left(L + \frac{1}{\omega^2}\right)}{\left(R^2 + \left(\omega L - \frac{1}{\omega}\right)^2\right)^2}$$

$$\frac{dV_L}{d\omega} = \frac{R^2 + \left(\omega L - \frac{1}{\omega c}\right)^2 (2\omega L^2 V^2) - (\omega^2 L^2 V^2)\left(2\left(\omega L - \frac{1}{\omega c}\right)\right)(}{\left(R^2 + \left(\omega c - \frac{1}{\omega}\right)^2\right)^2}$$

$$0 = V^2 \left[R^2 + \left(\omega_L - \frac{1}{\omega c}\right)^2 (L^2 2\omega) - \omega^2 L^2\, 2\left(\omega L - \frac{1}{\omega^2 c}\right)\left(L + \frac{1}{\omega^2 c}\right)\right]$$

$$\left(R^2 + \left(\omega L - \frac{1}{\omega c}\right)^2 \cancel{2}\cancel{L^2}\cancel{\omega} = \cancel{2}\cancel{L^2}\omega^{\cancel{2}} \left(\omega L - \frac{1}{\omega c}\right)\left(L + \frac{1}{\omega^2 c}\right)\right.$$

$$R^2 + \left(\frac{\omega^2 LC - 1}{\omega C}\right)^2 = \omega\left(\frac{\omega^2 LC + 1}{\omega^2 C}\right)$$

$$\frac{R^2\omega^2C^2 + (\omega^2LC - 1)^2}{\cancel{\omega^2} C^2} = \frac{\omega^3 LC + \omega}{\cancel{\omega^2}\,\cancel{C}}$$

$$R^2\omega^2C^2 + (\omega^2LC - 1)^2 = \omega^2LC^2 + \omega C$$

$$R^2\omega^2C^2 + (\omega^2LC - 1)^2 - \omega^2LC^2 - \omega C = 0$$

$$R^2\omega^2C^2 + 2 - 2\omega^2LC = 0$$

$$\cancel{\omega^2(R^2C^2 - 2LC)} = 0$$

$$\omega^2 = \frac{-2}{R^2C^2 - 2LC}$$

X -

$$\omega^2 = \frac{\cancel{-}2}{\cancel{-}(2LC - R^2C^2)}$$

$$\boxed{\omega = \sqrt{\frac{2}{2LC - R^2C^2}}}$$

$$\omega = \sqrt{\frac{\cancel{2}}{\cancel{2}LC\left(1 - \frac{R^2C^{\cancel{2}}}{2L\cancel{C}}\right)}}$$

$$\omega = \sqrt{\frac{1}{LC\left(1 - \frac{R^2C}{2L}\right)}}$$

WKT $\frac{1}{\sqrt{LC}} = \omega_0$

$$\boxed{\omega = \frac{\omega_0}{\sqrt{\left(1 - \frac{R^2C}{2L}\right)}}}$$

Parallel Resonance

Like series RLC ckts the resonance can be acheived in parallel RLC ckts, by varying frequency or LC components. in parallel RLC ckt the ckt behaves like a resistive network at Resonance. But resistance is maximum and current is minimum. hence it is called as anti-resonance circuits.

- for parallel we need to consider admitance.

$$V|Y| = I$$

- parallel resonant ckts is analysed in terms of admitance Y

Note:

series $|Z|$ = Reactance

parallel $|X|$ = susptance = B

In case of admitance B(L & C) ckts have opposite signs

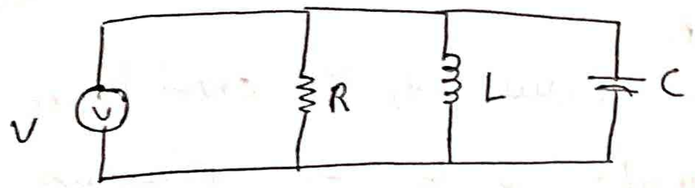
$$Y = \frac{1}{Z} = \frac{1}{R+jX} = G \pm jB$$

at particular frequency, under resonance, susptance of Inductance is equal to susptance of capacitor and hence its gets cancled.

$$Y = \frac{1}{R+jX}$$

$$Y = \frac{1}{R+j(x_c - X_c)}$$

Consider the ||le RLC ckt



The total impedance of the ckt

$$\frac{1}{Z} = \frac{1}{R} + \frac{1}{Z_L} + \frac{1}{Z_C}$$

$$Y = \frac{1}{R} + \frac{1}{jX_L} + \frac{1}{\frac{1}{-jX_C}}$$

$$= \frac{1}{R} - \frac{j}{\omega L} + \frac{1}{\frac{-j}{\omega C}}$$

$$Y = \frac{1}{R} - \frac{j}{\omega L} + \frac{j}{\frac{1}{\omega C}}$$

$$Y = \frac{1}{R} - \frac{j}{\omega L} + j\omega C$$

$$\boxed{Y = \frac{1}{R} + j\underbrace{\left(\omega C - \frac{1}{\omega L}\right)}_{B}} \quad \ldots \text{①}$$

at resonance

$$\boxed{Y = \frac{1}{R}} \quad \text{or} \quad \boxed{Y = G}$$

G = conductance

from ①

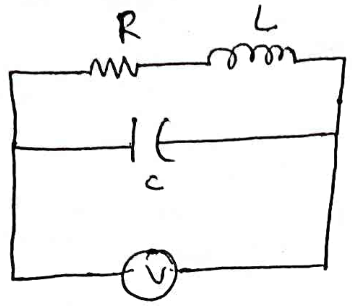
$$\boxed{\therefore Y = G + jB}$$

from eqn ① at resonance $X_L = X_C$

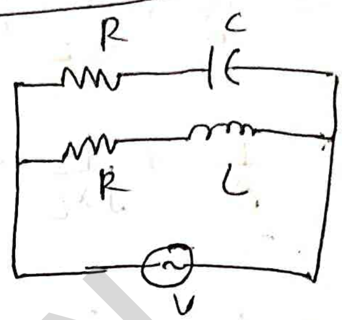
$$Y = \frac{1}{R}$$

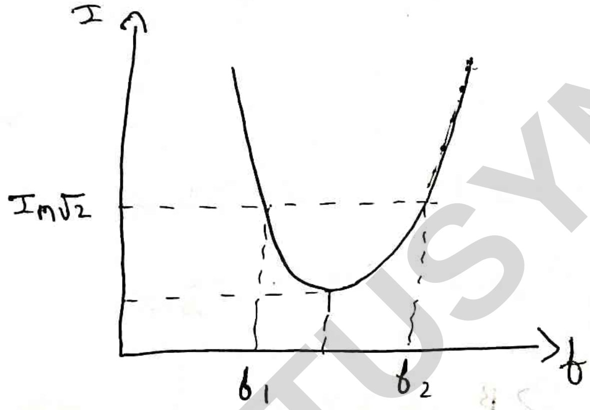
which is the minimum value of Y and hence magnitude of Z increases which make I to be minimum. (maximum)

→ case ②:



case ③:





→ Q factor:

~~diff~~ defined as ratio of = $\frac{\text{current through cap}}{\text{total current}}$

$$Q = \frac{I_{Cr}}{I_r} = \frac{\cancel{V}/X_{Cr}}{\cancel{V}/R}$$

$$Q = \frac{R}{X_{Cr}} = 2\pi f R C$$

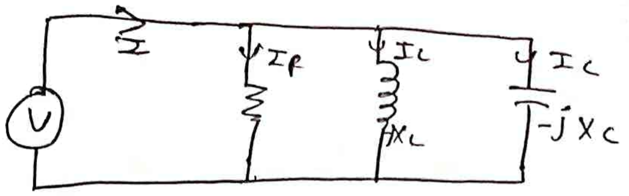
$$\boxed{Q_c = 2\pi f R C} \qquad \boxed{Q_c = \omega_0 R C}$$

$$\boxed{Q_L = \frac{R}{\omega_0 L}}$$

Expression for ω_1 & ω_2 (when RLC are parallel)

Case I

consider ||le RLC ckt



The Total current I is given by

$$I = I_R + I_L + I_C \quad \ldots \text{①}$$

But $I = \frac{V}{R}$

$$I = \frac{V}{R} + \frac{V}{jX_L} + \frac{V}{-jX_C}$$

$$I = \frac{V}{R} + \frac{V}{j\omega L} - \frac{V}{\frac{1}{j\omega C}}$$

$$I = V\left[\frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)\right]$$

$$I = VY$$

where $\boxed{Y = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)}$

$$|Y| = \sqrt{\frac{1}{|R|^2} + \left(\omega C - \frac{1}{\omega L}\right)^2} \quad \ldots \text{②}$$

$$|Y| = \frac{I}{V}$$

$$I = \frac{I_{min}\sqrt{2}}{\cancel{V}}$$

$$|Y| = \frac{I_{min}\sqrt{2}}{V}$$

$$|Y| = \frac{\frac{\cancel{V}}{R}\sqrt{2}}{\cancel{V}}$$

$$|Y| = \frac{\sqrt{2}}{R} \quad \ldots (3)$$

equating (2) & (3)

$$\frac{\sqrt{2}}{R} = \sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2}$$

squaring on both sides

$$\frac{2}{R^2} = \frac{1}{R^2} + \left(\omega_C - \frac{1}{\omega L}\right)^2$$

$$\frac{1}{R^2} = \left(\omega_C - \frac{1}{\omega L}\right)^2$$

$$\left(\omega C - \frac{1}{\omega L}\right)^{\cancel{2}} = \pm\frac{1}{R} \quad \ldots\ldots (4)$$

at $\omega = \omega_1$

$$\left(\omega_1 C - \frac{1}{\omega_1 L}\right) = -\frac{1}{R}$$

$$\frac{(\omega_1^2 CL - 1)}{\omega_1 L} = -\frac{1}{R}$$

$$R(\omega_1^2 CL - 1) = -\omega_1 L$$

$$\omega_1^2 RLC - R = -\omega_1 L$$

$$\omega_1^2 RLC + \omega_1 L - R = 0 \qquad \text{TF}$$

$$a = RLC \quad B = L \quad C = -R$$

$$x = \omega_1 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\omega_1 = \frac{-L \pm \sqrt{L^2 - 4(RLC)(-R)}}{2LC}$$

$$\omega_1 = \frac{-L \pm \sqrt{L^2 + 4R^2LC}}{2RLC}$$

$$\omega_1 = \frac{-L}{2RLC} \pm \sqrt{\frac{\cancel{L^2}}{4R^2\cancel{L^2}C^2} + \frac{\cancel{4}\cancel{R^2}\cancel{L}C}{\cancel{4}\cancel{R^2}L^2C^2}}$$

$$\omega_1 = \frac{-1}{2RC} \pm \sqrt{\frac{1}{4R^2C^2} + \frac{1}{LC}}$$

$$\omega_1 = \frac{-1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

at $\omega = \omega_2$

$$\omega_2 C - \frac{1}{\omega_2 L} = \frac{1}{R}$$

$$R(\omega_2^2 CL - 1) = \omega_2 L$$

$$\omega_2^2 RCL - R - \omega_2 L = 0$$

$$a = RCL \quad B = -L \quad C = -R$$

$a = RLC \quad B = -L \quad C = -R$

$$X = \omega_2 = \frac{L \pm \sqrt{L^2 + 4R^2LC}}{2RLC}$$

$$X = \omega_2 = \frac{\cancel{L}}{2R\cancel{L}C} \pm \sqrt{\frac{L^2 + 4R^2LC}{4R^2L^2C^2}}$$

$$X = \omega_2 = \frac{1}{2RC} \pm \sqrt{\frac{\cancel{L^2}}{4R^2\cancel{L^2}C^2} + \frac{\cancel{4R^2}\cancel{L}C}{\cancel{4R^2}L^{\cancel{2}}C^{\cancel{2}}}}$$

$$\boxed{\omega_2 = \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}}$$

$BW = f_2 - f_1$

$BW = \omega_2 - \omega_1$

$$BW = \frac{1}{2RC} + \cancel{\sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}}^{\,0} + \frac{1}{2RC} - \cancel{\sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}}$$

$$BW = \frac{1}{2RC} + \frac{1}{2RC}$$

$$BW = \frac{1}{2RC}(1+1)$$

$$BW = \frac{\cancel{2}}{\cancel{2}RC}$$

$$\boxed{BW = \frac{1}{RC}} \; //$$

$$\boxed{BW = \frac{1}{RC2\pi} \text{ Hz}}$$

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→ Expressing ω_1 & ω_2 in terms of BW

$$\omega_1 = -\frac{BW}{2} \pm \sqrt{\left(\frac{B}{2}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{BW}{2} \pm \sqrt{\left(\frac{BW}{2}\right)^2 + \frac{1}{LC}}$$

Prove that parallel resonant frequency is geometric mean of half power frequency.

$$\omega_1 . \omega_2 = \left(-\frac{1}{2RC} + \sqrt{\frac{1}{2RC} + \frac{1}{LC}}\right) \cdot \left(\frac{1}{2RC} + \sqrt{\frac{1}{2RC} + \frac{1}{LC}}\right)$$

$$\omega_1 . \omega_2 = -\frac{1}{4R^2C^2} \cancel{-\left(\frac{1}{2RC}\left(\sqrt{\frac{1}{2RC}+\frac{1}{LC}}\right)\right)} \cancel{+\left(\frac{1}{2RC}\left(\sqrt{\frac{1}{2RC}+\frac{1}{LC}}\right)\right)}$$

$$\left(\sqrt{\frac{1}{2RC} + \frac{1}{LC}}\right)\left(\sqrt{\frac{1}{2RC} + \frac{1}{LC}}\right)$$

$$\omega_1 . \omega_2 = -\frac{1}{4R^2C^2} + \left(\sqrt{\frac{1}{2RC} + \frac{1}{LC}}\right)^2$$

$$\omega_1 - \omega_2 = \cancel{-\frac{1}{4R^2C^2}} + \cancel{\frac{1}{2RC}} + \frac{1}{LC}$$

$$\omega_1 - \omega_2 = \frac{1}{LC}$$

$$\omega_1 - \omega_2 = \omega_0^2$$

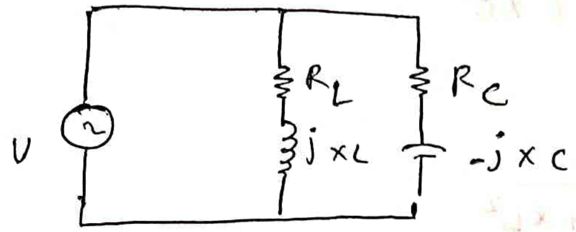
$$\omega_0 = \sqrt{\omega_1 . \omega_2}$$

$$\cancel{2\pi} f_0 = \cancel{2\pi}\sqrt{f_1 . f_2}$$

$$\boxed{f_0 = \sqrt{f_1 . f_2}}$$

CO2

• Derive an Exp for f_0 ll le Resonant circuit also show that the circuit will resonate at all frequency if $R_L = R_C = \sqrt{\frac{L}{C}}$ where R_L is the Resistor in inductor branch, R_C is the Resistor in capacitor branch.



Consider parallel resonant ckt as shown above

WKT, $Y = G + jB$

The total admitance given by

$$Y = Y_L + Y_C \quad \ldots (2)$$

$$Y_L = \frac{1}{Z_L} = \frac{1}{R_L + jX_L} \times \frac{R_L - jX_L}{R_L - jX_L}$$

$$Y_L = \frac{R_L - jX_L}{R_L^2 - (jX_L)^2} = \frac{R_L - jX_L}{R_L^2 + X_L^2} \quad \ldots (3)$$

$$Y_C = \frac{1}{Z_C} = \frac{1}{R_C - jX_C} \times \frac{R_C + jX_C}{R_C + jX_C}$$

$$Y_C = \frac{1}{Z_C} = \frac{R_C + jX_C}{R_C^2 + X_C^2} \quad \ldots (4)$$

Subs (3) & (4) in (2)

$$Y = \frac{R_L - jX_L}{R_L^2 + X_L^2} + \frac{R_C + jX_C}{R_C^2 + X_C^2}$$

$$Y = \frac{R_L}{R_L^2 + X_L^2} - \frac{jX_L}{R_L^2 + X_L^2} + \frac{R_C}{R_C^2 + X_C^2} + \frac{jX_C}{R_C^2 + X_C^2}$$

$$Y = \frac{R_L}{R_L^2 + X_L^2} + \frac{R_C}{R_C^2 + X_C^2} + j\left(\frac{X_C}{R_C^2 + X_C^2} - \frac{X_L}{R_L^2 + X_L^2}\right)$$

Comparing (5) & (1)

$$G = \frac{R_L}{R_L^2 + X_L^2} + \frac{R_C}{R_C^2 + X_C^2}$$

$$B = \frac{X_C}{R_C^2 + X_C^2} - \frac{X_L}{R_L^2 + X_L^2}$$

at resonance $B_L = B_C$

$$B_L - B_C = 0$$

$$B = 0$$

$$\frac{X_C}{R_C^2 + X_C^2} - \frac{X_L}{R_L^2 + X_L^2} = 0$$

$$\frac{X_C}{R_C^2 + X_C^2} = \frac{X_L}{R_L^2 + X_L^2}$$

X_C as $\frac{1}{\omega C}$ $\qquad X_L = \omega L$

$$\frac{1}{\omega C(R_C^2 \times \frac{1}{\omega C})^2} = \frac{\omega L}{R_L^2 + X_L^2}$$

$$\cancel{R_L^2 + X_L^2 = (\omega^2 C^2 L)(\omega^2 R_L C^3)}$$

$$\cancel{\omega^2 C^2 (}$$

$$\cancel{R_L^2 + X_L^2 = (\omega^2 C^2 L)(}$$

(c)

$$\frac{\cancel{\omega} C}{R_C^2\omega^2C^2+1} = \frac{\cancel{\omega} L}{R_L^2+\omega L^2}$$

$$R_L^2 C + \omega_L^2 C = R_C^2\omega^2 L C^2 + L$$

$$\omega^2 = \frac{L - CR_L^2}{L^2C - LC^2R_C^2}$$

$$\omega = \sqrt{\frac{L - CR_L^2}{L^2C - LC^2R_C^2}}$$

at Resonance

$$Y = \frac{R_L}{R_L^2 + X_L^2} + \frac{R_C}{R_C^2 + X_C^2}$$

$$Y = \frac{R_L(R_C^2 + X_C^2) + R_C(R_L^2 + X_L^2)}{(R_L^2 + X_L^2)(R_C^2 + X_C^2)}$$

$$Y = \frac{R_L R_C^2 + R_L X_C^2 + R_C R_L^2 + R_C X_L^2}{R_L^2 R_C^2 + R_L^2 X_C^2 + R_C^2 X_L^2 + X_L^2 X_C^2}$$

Assume : $R_L = R_C = R$, $\boxed{X_L^2 \times X_C^2 = R^4}$

$$Y = \frac{R^3 + R X_C^2 + R^3 + R X_L^2}{R^4 + R^2 X_C^2 + R X_L^2 + X_L^2 X_C^2} \quad X_L^2 X_C^2 \rightarrow R^4$$

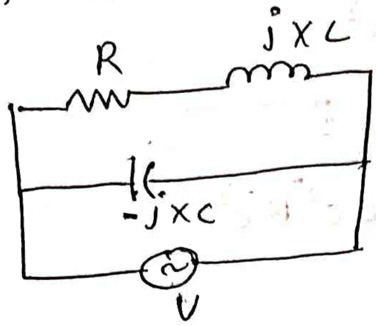
$$Y = \frac{2R^3 + R X_C^2 + R X_L^2}{2R^4 + R^2 X_C^2 + R^2 X_L^2}$$

$$Y = \frac{R\,\cancel{[2R^2 + X_C^2 + X_L]}}{R^2\,\cancel{[2R^2 + X_C^2 + X_L]}} \qquad Y = \frac{1}{R}$$

$$Y = \frac{1}{R} = \frac{1}{\sqrt{\frac{L}{C}}}$$

$$\boxed{Y = \sqrt{\frac{C}{L}}}$$

Case 3: consider the network



$$Y = G + jB \quad \ldots (1)$$

The total admittance is given by

$$Y = Y_L + Y_C \quad \ldots (2)$$

$$Y_L = \frac{1}{Z_L} = \frac{1}{R + jX_L} \times \frac{R_L - jX_L}{R_L - jX_L}$$

$$Y_L = \frac{R_L - jX_L}{R_L^2 - (jX_L)^2} = \frac{R_L - jX_L}{R_L^2 + (X_L)^2} \quad \ldots (3)$$

$$Y_C = \frac{1}{Z_C} = \frac{1}{-jX_C} = \frac{j}{X_C} \quad \ldots (4)$$

Sub (3) & (4) in (1)

$$Y = \frac{R_L - jX_L}{R_L^2 + X_L^2} + \frac{j}{X_C}$$

$$Y = \frac{R_L}{R_L^2 + X_L^2} - \frac{jX_L}{R_L^2 + X_L^2} + \frac{j}{X_C}$$

$$Y = \frac{R_L}{R_L^2 + X_L^2} + j\left(\frac{1}{X_C} - \frac{X_L}{R_L^2 + X_L^2}\right) \quad \ldots (5)$$

comparing 5 & 1

$$G = \frac{RL}{RL^2 + XL^2}$$

$$B = \frac{1}{xc} - \frac{XL}{R_L^2 + XL^2}$$

at resonance $B = 0$

$$0 = \frac{1}{x_c} - \frac{X_L}{R_L^2 + X_L^2}$$

$$\frac{X_L}{R_L^2 + X_L^2} = \frac{1}{X_C}$$

$$X_L X_C = R_L^2 + X_L^2$$

$$X_L X_C - R_L^2 - X_L^2$$

$$\cancel{\omega} L \cdot \frac{1}{\cancel{\omega} C} - R_L^2 - (\omega L)^2 = 0$$

$$\frac{L}{C} - R_L^2 - \omega^2 L^2 = 0$$

$$\frac{L}{C} - R_L^2 = \omega^2 L^2$$

$$\omega^2 = \frac{\frac{L}{C} - R_L^2}{L^2}$$

$$\omega^2 = \frac{\cancel{L}}{L^2 C} - \frac{R_L^2 \cancel{C}}{\cancel{L^2 C}} = \frac{1}{LC} - \frac{R_L^2}{L}$$

$$\boxed{\omega = \sqrt{\frac{1}{LC} - \left(\frac{R_L}{L}\right)^2}}$$

at resonat $B = 0$

eq ① $= y = G + jB$

$$Y = G$$

$$Y = \frac{R_L}{R_L^2 + X_L^2}$$

$$Y = \frac{R_L}{R_L^2 + \omega^2 L^2}$$

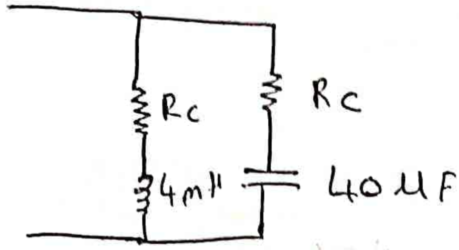
$$Y = \frac{R_L}{R_L^2 + \left(\sqrt{\frac{1}{LC} - \left(\frac{R_L}{L}\right)^2}\right)^2 L^2}$$

$$Y = \frac{R_L}{R_L^2 + \left(\frac{1}{LC} - \left(\frac{R_L}{L}\right)^2\right) L^2}$$

$$Y = \frac{R_L}{\cancel{R_L^2} + \frac{L}{C} - \cancel{R_L^2}}$$

$$\boxed{Y = \frac{R_L C}{L}}$$

-> Find the value of R_C and R_C for fig shown resonate at all frequency the ckt can resonate at any frequency if $R_L^2 = R_C^2 = \frac{L}{C}$

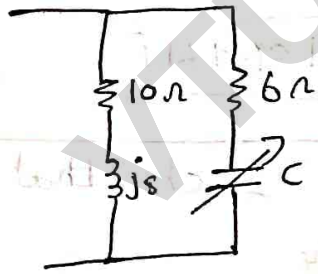


WKT $R_L = R_C = \sqrt{\frac{L}{C}}$

$$\sqrt{\frac{L}{C}} = \sqrt{\frac{4m}{40\mu}} = 10\Omega$$

$$\boxed{\therefore \quad R_L = R_C = 10\Omega}$$

-> Find the value of c for which ckt given in fig resonates at 750 Hz



$$y = y_L + y_C$$

$$y_L = \frac{1}{Z_L} = \frac{1}{10+j8} = 0.06 - 0.04j$$

$$y_C = \frac{1}{Z_C} = \frac{1}{6 - jX_C} = \frac{6 + jX_C}{6 + jX_C}$$

$$y_C = \frac{6 + jX_C}{6^2 - (jX_C)^2} = \frac{6 + jX_C}{36 + X_C^2}$$

$$y = \left(\frac{10}{164} + \frac{6}{36 + (x_c)^2}\right) + j\left(\frac{x_c}{36 + x_c^2} - \frac{8}{164}\right)$$

at resonance

$$B = 0$$

$$y = \frac{x_c}{36 + x_c^2} = \frac{8}{164}$$

$$y = 164\, x_c = 8\, x_c^2 + \cancel{164}\,288$$

$$8x_c^2 - 164x_c + 288 = 0$$

$$\boxed{x_{c_1} = 18.56\,\Omega} \qquad \boxed{X_{c_2} = 1.93\,\Omega}$$

$$C_1 = \frac{1}{\omega x_{c_1}} \qquad C_2 = \frac{1}{\omega X_{c_2}}$$

$$C_1 = \frac{1}{2\pi(750)\,18.56} \qquad C_2 = \frac{1}{2\pi(750)\,1.93}$$

$$\boxed{C_1 = 11.43\,\mu F} \qquad \boxed{C_2 = 109.9\,\mu F}$$

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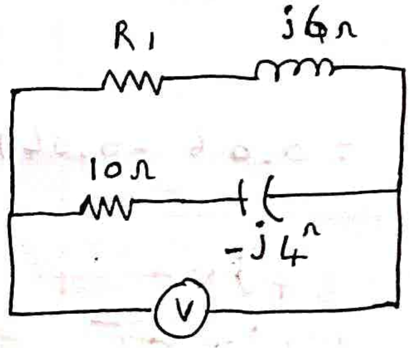
3) Determine the value of R_1 such that the ckt given in fig resonates



$$Y = Y_L + Y_C$$

$$X_L = \frac{1}{Z_L} = \frac{1}{R_1 + j6} \cdot \frac{R_1 - j6}{R_1 - j6}$$

$$\boxed{Y_L = \frac{R_1 - j6}{R_1^2 + 36}}$$

$$Y_C = \frac{1}{Z_C} = \frac{1}{10 - j4} \times \frac{10 + j4}{10 + j4}$$

$$Y_C = \frac{10 + j4}{100 + 16} \qquad Y_C = \frac{10 + j4}{116}$$

$$\therefore Y = \left(\frac{R_1 - j6}{R_1^2 + 36}\right) + \left(\frac{10 + j4}{116}\right)$$

$$Y = G + jB$$

$$Y = \left(\frac{R_1}{R_1^2 + 36} + \frac{10}{116}\right) + j\left(\frac{4}{116} - \frac{6j}{R_1^2 + 36}\right)$$

at resonance $B = 0$

$$\cancel{Y} = B = \frac{4}{116} - \frac{6\cancel{j}}{R_1^2 + 36}$$

$$\frac{6\cancel{j}}{R_1^2 + 36} = \frac{4}{116}$$

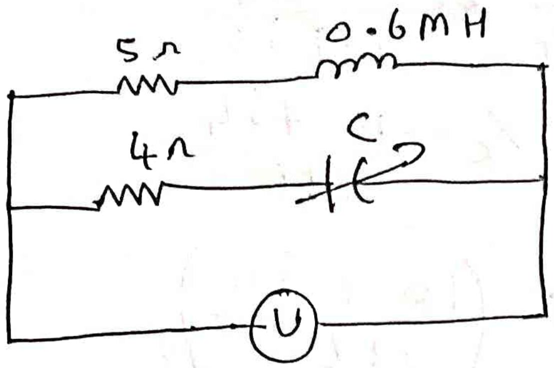
$$696 = 4R_1^2 + 144$$

$$R_1^2 = \frac{696 - 144}{4}$$

$$R_1 = \sqrt{138}$$

$$\boxed{R_1 = 11.74\,\Omega}$$

→ for the ckt shown find two values of C when f of driving voltage is 5000 R/s



$C_1 = 20.66\ \mu F$

$C_2 = 121.06\ \mu F$

$$Y = Y_L + Y_C$$

$$Y_L = \frac{1}{Z_L} = \frac{1}{5 + j(0.6 \times 10^{-3})} \cdot \frac{5 - j(0.6 \times 10^{-3})}{5 - j(0.6 \times 10^{-3})}$$

$$Y_L = \frac{5 - j(0.6 \times 10^{-3})}{25 + (0.36 \times 10^{-6})}$$

$$Y_C = \frac{1}{Z_C} = \frac{1}{4 - jX_C} \cdot \frac{4 + jX_C}{4 + jX_C} = \frac{4 + jX_C}{16 + (X_C)^2}$$

$$Y = \frac{5 - j(0.6 \times 10^{-3})}{25 + (0.36 \times 10^{-6})} + \frac{4 + jX_C}{16 + (X_C)^2}$$

$$Y = \left(\frac{5}{25 + 0.36 \times 10^{-6}} + \frac{4}{16 + (X_C)^2}\right) + j\left(\frac{X_C}{16 + X_C^2} - \frac{0.6 \times 10^{-3}}{25.36 \ldots}\right)$$

at resonance $B = 0$

$$\frac{X_C}{16 + X_C^2} = \frac{0.6 \times 10^{-3}}{25 + 36 \times 10^{-6}}$$

$$0.6 \times 10^{-3} X_C^2 - 25\cancel{.36 \times 10^{-6}}\, X_C + 9.6 \times 10^{-3} = 0$$

$$\boxed{X_{C2} = 0.02 - 3.9 i}$$

$X_{C1} = 4166.66$

$X_{C2} = 3.84 \times 10^{-4}$

$$C_1 = \frac{1}{\omega X_{C1}} =$$

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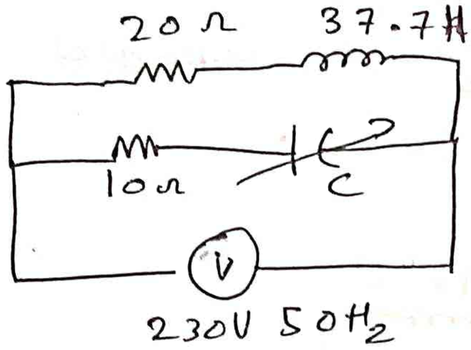
→ Find the value of C for ckt shown.



$C_1 = 68.9\,\mu F$

$C_2 = 1.46\, mF$

$$y = y_L + y_C$$

$$y_L = \frac{1}{20 + j(37.7)} \times \frac{20 - j(37.7)}{20 - j(37.7)} \qquad y_L = \frac{20 - j(37.7)}{400 + 1421.2}$$

$$\boxed{y_L = \frac{20 - j(37.7)}{400 + 1421.29}}$$

$$y_C = \frac{1}{10 - jX_C} \quad \frac{10 + jX_C}{10 + jX_C} = \frac{10 + jx_C}{100 + x_C^2}$$

$$\therefore y = \left(\frac{20}{1821.29} + \frac{10}{100 + x_C^2}\right) + j\left(\frac{x_C}{100 + x_C^2} - \frac{37.7}{1821.29}\right)$$

$\therefore$

$$\frac{x_C}{100 + x_C^2} = \frac{37.71}{1821.29}$$

10^{-6}

$$1821.29\, X_C = 3771 + 37.71\, x_C^2$$

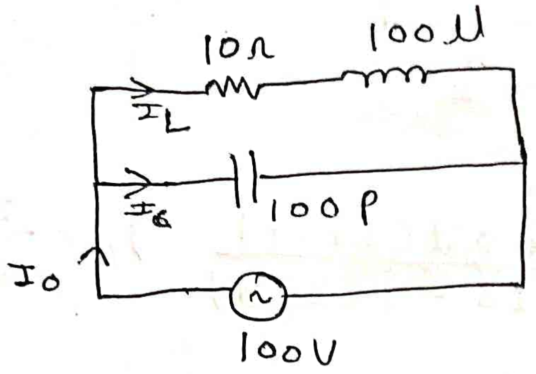
$$-37.71\, x_C^2 + 1821.29\, x_C - 3771 = 0$$

$x_{C1} = 46.129$ $\qquad$ $x_{C2} = 2.1678$

$$C_1 = \frac{1}{2\pi f\, x_1} \qquad C_2 = \frac{1}{2\pi f\, X_{C2}}$$

$\boxed{C_1 = 69.004\,\mu F}$ $\qquad$ $\boxed{C_2 = 1.46\, mF}$

For the ||le resonant ckt shown find I_L, I_o, I_c, f_o, and Dynamic impedance for the ckt shown



$$Z_o = \frac{L}{RC}$$

$$f_o = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \left(\frac{R_L}{L}\right)^2}$$

$$\boxed{f_o = 1.591 \text{ MHz}}$$

$$I_o = \frac{V}{Z_o}$$

$$I_o = \frac{100}{100K}$$

$$\boxed{I_o = 1 mA}$$

$$I_L = \frac{100}{R + jX_L}$$

$$I_L = \frac{100}{10 + j(2\pi f(100\mu))}$$

$$\boxed{I_L = 1.00 \times 10^{-3} - 0.1i}$$

$$I_c = \frac{V}{X_C}$$

$$I_C = \frac{100}{\frac{1}{j\omega C}} = j\,100\,\omega C$$

$$I_C = j(100 \times 2\pi f \times 100 \times 10^{-12})$$

$$\boxed{I_C = 0.099\,j}$$

→ In a ∥le RLC, $C = 50\,\mu F$ determine BW, Q, L and R

$$\omega_0 = \sqrt{\omega_1 \omega_2}$$

$$\omega_0^2 = \omega_1 \cdot \omega_2$$

$$BW = \frac{\omega_0}{Q}$$

case (i) $\omega_0 = 100$, $\omega_2 = 120$

(ii) $\omega_0 = 100$, $\omega_1 = 80$

soln. $\omega_0^2 = \omega_1 \cdot \omega_2$

$$\omega_1 = \frac{\omega_0^2}{\omega_2} = \frac{100^2}{120} = 83.3 \text{ rad/s}$$

$$BW = \omega_2 - \omega_1$$

$$\boxed{BW = 36.7 \text{ rad/s}}$$

Q fad = $B.W = \frac{\omega_0}{Q}$

$$Q = \frac{\omega_0}{BW}$$

$$\boxed{Q = 2.724}$$

$$Q = \omega_0 R C$$

$$R = \frac{Q}{\omega_0 C} = 544\,\Omega$$

$$q = \frac{R}{\omega_0 L}$$

$$L = \frac{R}{\omega_0 q}$$

$$\boxed{L = 2H}$$

case(2): $\omega_0 = 100$ | $\omega_1 = 80$

$$\omega_0^2 = \omega_1 \cdot \omega_2$$

$$\omega_2 = \frac{\omega_0^2}{\omega_1}$$

$$\omega_2 = 125 \text{ Rad/s}$$

$$BW = \omega_2 - \omega_1$$

$$BW = 125 - 80$$

$$\boxed{BW = 45 \text{ Rad/s}}$$

$$q = \frac{\omega_0}{BW}$$

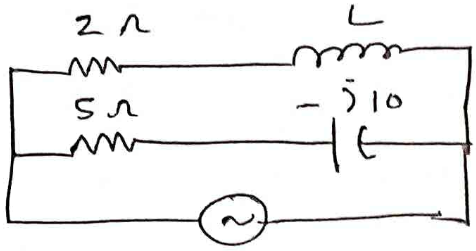
$$\boxed{q = 2.22}$$

$$q = \omega_0 RC$$

$$R = \frac{q}{\omega_0 C} = 444\,\Omega$$

$$L = \frac{R}{\omega_0 q} = 2H$$

→ Find the value of (L) at which ckt resonates ω = 500 rad/sec



$$y = y_L + y_C$$

$$y_L = \frac{1}{Z_L} = \frac{1}{2+jX_L} = \frac{1}{2+jX_L} \cdot \frac{2-jX_L}{2-jX_L}$$

$$y_L = \frac{2-jX_L}{4+X_L^2}$$

$$y_C = \frac{1}{Z_C} = \frac{1}{5-j10} = \frac{1}{5-j10}\left(\frac{5+j10}{5+j10}\right)$$

$$y_C = \frac{5+j10}{25+100}$$

$$y_C = \frac{5+j10}{125}$$

$$y = \frac{2-jX_L}{4+X_L^2} + \frac{5+j10}{125}$$

$$y = \frac{2}{4+X_L^2} + \frac{5}{125} + j\left(\frac{10}{125} - \frac{X_L}{4+X_L^2}\right)$$

at resonance $B = 0$

$$\frac{10}{125} = \frac{X_L}{4+X_L^2}$$

$$40 + 10X_L^2 = 125X_L$$

$$10X_L^2 + 40 - 125X_L = 0$$

$X_{L1} = 12.17$ $\quad$ $X_{L2} = 0.3288$

$$x_L = 2\pi f L$$

$$x_L = \omega L$$

$$L_1 = \frac{x_{L_1}}{\omega} = \frac{12.17}{500} = 24.34 \text{ mH}$$

$$L_2 = \frac{X_{L_2}}{\omega} = \frac{0.3286}{500} = 0.657 \text{ mH}$$